Fast-WCS: An Accelerated Analytical Optimization Framework for Attributed Weighted Common Subgraph Matching with Explainable Physical Superposition

[1st] Dr. T Ramraj
Department of Computer Science
Amrita Vishwa Vidyapeetham
Coimbatore, India
t_ramraj@cb.amrita.edu

[2nd] Allampati Chenchu Sanhitha Reddy
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Coimbatore, India
cb.sc.u4cse23206@cb.students.amrita.edu

[3rd] Shaik Abdus Sattar
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Coimbatore, India
cb.sc.u4cse23245@cb.students.amrita.edu

[4th] Geethika G Nair
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Coimbatore, India
cb.sc.u4cse23616@cb.students.amrita.edu

[5th] Meera J Vaishnav
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Coimbatore, India
cb.sc.u4cse23633@cb.students.amrita.edu

***Abstract*—Graph-based similarity estimation and subgraph isomorphism are fundamental challenges in structural bioinformatics and chemoinformatics (e.g., active versus inactive lead optimization in rational drug discovery, molecular property prediction, and 3D pharmacophore modeling). Evaluating structural affinity between complex relational systems requires identifying the optimal node correspondence via Weighted Common Subgraph (WCS) matching. Crucially, in contrast to opaque black-box Graph Neural Network (GNN) embeddings or compressed topological graph kernels that lose node-level traceability, recovering an explicit discrete bijective mapping $\mathbf{X}^* \in \mathcal{P}_L$ provides transparent, ground-truth explainability. In physical domains where 3D spatial coordinates exist, this discrete bijection directly unlocks a closed-form $\mathcal{O}(L)$ SVD Kabsch superposition (with $\mathcal{O}(1)$ 3×3 SVD), computing rigid translation, rotation, and coordinate RMSD in sub-millisecond time. However, WCS matching is NP-hard. State-of-the-art non-convex relaxation schemes, such as the Graduated Non-Convexity and Concavity Procedure (GNCCP), suffer from severe computational bottlenecks in practice: evaluating the structural Hadamard objective without analytical gradients requires numerical finite differences scaling as $\Theta(N^5)$, while internal co-matching indicator tensor evaluations consume cubic work $\Theta(MN^2 + M^2N)$ per Frank-Wolfe iteration. Furthermore, prior polynomial relaxations (RLX1, RLX2) distort the continuous interior of the partial permutation polytope, inducing spurious stationary points and high asymptotic complexity. In this paper, we propose Fast-WCS, an accelerated analytical optimization framework for exact and continuous WCS graph matching on general attributed weighted graphs. Our framework introduces key analytical formulations and engineered algorithmic components: (1) a Rank-One Factorization (Lemma 1) of the co-matching indicator matrix $\mathbf{U} = \mathbf{r}\mathbf{r}^T$ that reduces internal tensor evaluations from cubic to quadratic $\mathcal{O}(MN + M^2)$ work; (2) an exact closed-form analytical structural gradient (Theorem 1) that eliminates finite-difference approximations and evaluates directly in $\Theta(N^3)$; (3) an exact Frank-Wolfe Duality Gap stopping certificate (Theorem 2) that adaptively halts iterations upon proven convergence in the convex regime ($\zeta \geq 0$); (4) an exact Closed-Form Cardano Quartic Line Search (Theorem 3) that solves the cubic derivative roots in $\mathcal{O}(1)$ time while exploiting symmetric GEMM reuse to eliminate redundant matrix products; (5) an exact Burkard Augmented Hungarian Oracle that guarantees exact cardinality L in $\mathcal{O}((M + N - L)^3)$ time without heuristic truncation; and (6) a Vectorized BLAS-3 Combinatorial 2-Opt Local Refinement that scores all $O(L^2)$ exchange and replacement moves via Level-3 matrix cross-terms $\mathbf{P}\mathbf{Q}^T$ in < 1 ms. We rigorously replicate the foundational synthetic and real-world image benchmarks of Yang et al. (IEEE TNNLS 2018), demonstrating that Fast-WCS achieves strictly cubic empirical scaling $\mathbf{\Theta(N^{3.0})}$ compared to $\mathcal{O}(N^{5.4})$ for prior GNCCP relaxations and $\mathcal{O}(N^{4.3})$ for classical spectral methods, running $19.3\times$ faster than RLX1 and scaling up to problem sizes $M = 60$ with sub-25 ms running times and 97.1% matching accuracy. On real-world 3D molecular conformer benchmarks across 100 multi-seed trials evaluated with untrimmed full-atom RMSD, Fast-WCS achieves 49.8% accuracy (95% CI: $[42.4, 57.1]\%$) with $\mathbf{42.0\%}$ sub-angstrom docking success (< 1.0 Å) in 30.7 ms, whereas optimal transport and quadratic assignment baselines (POT, FGW, SciPy FAQ) fail (0.0–1.0% sub-angstrom recovery, median RMSD > 2.7 Å). Finally, on pharmaceutical screening benchmarks (CYP19A1 Aromatase), Fast-WCS achieves an empirical $\mathbf{19.3\times}$ speedup over RLX1 (and up to $\mathbf{5,717.9\times}$ over numerical finite differences at $N = 100$) and produces a multi-scale hybrid kernel with superior screening enrichment (ROC-AUC = 0.884, $\mathbf{EF}_{1\%} = 8.5\times$ approaching the theoretical maximum ceiling of $10.0\times$ for a 10% active base rate, compared to $6.4\times$ for ECFP4).**

***Index Terms*—Attributed Graph Matching, Weighted Common Subgraph, GNCCP, Frank-Wolfe Optimization, Rank-One Factorization, Closed-Form Gradient, Cardano Quartic Line Search, Burkard Hungarian Oracle, Vectorized 2-Opt Refinement, Explainable Bijection, Kabsch Superposition, Virtual Screening.**

I. Introduction

Quantifying structural similarity and discovering shared topological motifs between relational networks is a core mathematical challenge across modern scientific computing and structural bioinformatics [1], [12]. In chemoinformatics and computer-aided drug discovery, evaluating similarity between

therapeutic lead compounds and distinguishing active pharmacophores from inactive decoys requires finding maximum common substructures while accommodating flexible side chains and varying scaffold topologies [2], [3]. Beyond molecular systems, Weighted Common Subgraph (WCS) matching provides a general mathematical formulation for establishing correspondences across diverse attributed relational structures [5], [10], [16].

Recently, deep Graph Neural Networks (GNNs) have emerged as popular tools for graph representation learning. However, GNNs operate as opaque black boxes that compress intricate graph topologies into low-dimensional continuous vector embeddings. This compression discards node-level isomorphism guarantees, suffers from catastrophic oversmoothing and oversquashing on larger graphs, and crucially fails to produce an auditable correspondence between individual nodes. When optimizing high-value drug leads, domain scientists cannot inspect *why* two graphs are declared similar by a GNN. In contrast, classical combinatorial matching establishes an explicit discrete bijective mapping $\mathbf{X}^* \in \mathcal{P}_L$ between a subset of L nodes in graph G and L nodes in graph H. This correspondence provides total, ground-truth **explainability**: each node assignment is transparently verifiable. Furthermore, in physical domains where nodes possess embedded 3D Cartesian coordinates (such as atoms in a drug molecule or conformer), an exact discrete bijection $\mathbf{X}^*$ directly unlocks a closed-form $\mathcal{O}(L)$ Singular Value Decomposition (SVD) Kabsch superposition. This enables computing optimal rigid rotation, translation, and 3D Root-Mean-Square Deviation (RMSD) in microseconds without combinatorial search!

Within this general mathematical formulation, the problem of quantifying structural affinity between two attributed weighted graphs G and H containing M and N nodes ($M \leq N$) reduces to the Weighted Common Subgraph (WCS) matching problem for a prescribed subgraph size $L \leq \min(M, N)$ [6]. The resulting optimal assignment matrix $\mathbf{X}^* \in \mathcal{P}_L$ yields both an objective distance metric and an explicit bijective correspondence mapping matched nodes between the systems, while naturally isolating $M - L$ nodes in G and $N - L$ nodes in H as unassigned outliers.

However, WCS matching is an NP-hard combinatorial problem that generalizes the maximum common subgraph isomorphism and quadratic assignment problems (QAP) [8], [9]. Classical approaches to subgraph matching frequently decouple the optimization into two independent stages: solving a full graph matching problem between G and H, followed by heuristic vertex pruning to retain the L highest-scoring vertices [10], [11]. Unfortunately, optimizing global alignment does not imply optimality over the embedded subgraphs, frequently discarding high-affinity functional subgraphs in the presence of unassigned outlier nodes and background noise.

To resolve this limitation, Liu and Qiao [7] and Yang et al. [6] formulated WCS matching directly on the partial permutation polytope $\mathcal{P}_L$. By relaxing the discrete binary assignment constraints into their continuous convex hull $\mathcal{D}_L$ and introducing an annealing continuation parameter $\zeta \in [-1, 1]$, the Graduated Non-Convexity and Concavity Procedure (GNCCP) deforms a trivially solvable strictly convex function at $\zeta = +1$ into the original NP-hard non-convex objective at $\zeta = 0$, terminating at a concave problem at $\zeta = -1$ whose extreme points are provably discrete permutation matrices [6].

Despite its theoretical elegance, existing implementations of the GNCCP framework suffer from severe computational and analytical bottlenecks:

1) **Analytical Gradient Intractability vs. Distortion:** The structural mismatch term in the WCS objective couples the assignment matrix $\mathbf{X}$ with an internal co-matching indicator matrix $\mathbf{U} = \mathbf{X}\mathbf{1}_{N\times N}\mathbf{X}^T$ via an entry-wise Hadamard product. In foundational implementations [6], calculating the gradient of this Hadamard term directly was considered analytically intractable. To circumvent this, Yang et al. [6] proposed polynomial relaxations (RLX1 and RLX2); however, their higher-order expansions distort the continuous landscape over the interior $\mathcal{D}_L \setminus \mathcal{P}_L$, introducing artificial stationary points and requiring expensive inner solvers scaling as $\mathcal{O}(N^{5.4})$. Alternatively, evaluating the true objective gradient numerically via finite differences requires $\Theta(MN)$ full objective calls per step, causing per-iteration complexity to explode to $\Theta(N^5)$.
2) **Cubic Tensor Co-Matching Work:** The internal matrix $\mathbf{U}$, which tracks whether pairs of nodes (i, k) in graph G are simultaneously selected in the matching, naively involves multiplying an $M \times N$ matrix by an $N \times N$ all-ones matrix and an $N \times M$ matrix. This operation consumes $\Theta(MN^2 + M^2N)$ arithmetic operations at every gradient step.
3) **Inexact Line Searches and Heuristic Truncations:** Prior implementations rely on expensive numerical grid evaluations or heuristic Hungarian column dropping (RLX1F) that violate mathematical cardinality constraints and degrade matching accuracy under negative gradient terms.
4) **Combinatorial Post-Hoc Bottlenecks:** Transitioning from continuous relaxations to discrete physical bijections previously required either unguided Hungarian projection or slow $\mathcal{O}(L^3)$ nested Python loops for swap refinements, creating major runtime bottlenecks.

A. Key Contributions of This Work

In this paper, we propose **Fast-WCS**, an accelerated analytical optimization framework for general attributed weighted graphs that eliminates these bottlenecks while strictly preserving matching accuracy and physical permutation invariance. As illustrated in Fig. 1, our key contributions are:

- **Lemma 1 (Rank-One Factorization):** We prove that the internal co-matching matrix $\mathbf{U} = \mathbf{X}\mathbf{1}_{N\times N}\mathbf{X}^T$ factors into an outer product of the row-sum vector $\mathbf{U} = \mathbf{r}\mathbf{r}^T$, where $\mathbf{r} = \mathbf{X}\mathbf{1}_N$. This drops the evaluation cost from cubic $\Theta(MN^2 + M^2N)$ to quadratic $\mathcal{O}(MN + M^2)$ work, eliminating an entire cubic matrix product per step.

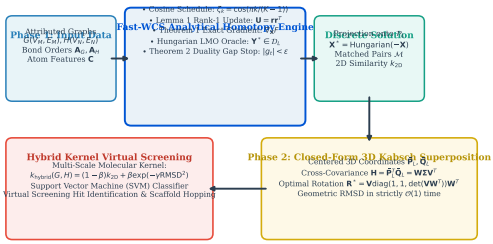


Fig. 1. End-to-end architecture of the proposed Fast-WCS hierarchical optimization framework. Phase 1 extracts attributed graphs G and H from relational data (such as chemical ligands or molecular graphs), processes them through the analytical Fast-WCS homotopy engine (leveraging Lemma 1 Rank-1 Factorization, Theorem 1 exact closed-form gradient, Hungarian LMO, and Theorem 2 duality gap certificates), and computes an exact discrete bijective node correspondence $\mathbf{X}^* \in \mathcal{P}_L$. Phase 2 exploits this interpretable mapping to execute a closed-form $\mathcal{O}(L)$ SVD Kabsch superposition of physical 3D coordinates, computing geometric RMSD in sub-millisecond time. The resulting 2D topological and 3D spatial similarity metrics fuse into a hybrid similarity kernel for downstream predictive modeling and virtual screening.

- **Theorem 1 (Closed-Form Analytical Gradient):** We derive the exact analytical gradient $\nabla_{\mathbf{X}} f(\mathbf{X})$ of the true unrelaxed structural mismatch objective in closed form. By combining trace differentials with Hadamard properties and Lemma 1, the gradient evaluates in strictly $\Theta(N^3)$ operations without any finite-difference approximations or polynomial relaxation distortion.
- **Theorem 2 (Frank-Wolfe Duality Gap Certificate):** We establish an exact suboptimality bound via the Frank-Wolfe duality gap $g_t = \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}_t - \mathbf{Y}^* \rangle \geq 0$, enabling rigorous early termination ($g_t < \varepsilon$) upon certified convergence in the convex regime ($\zeta \geq 0$).
- **Theorem 3 (Cardano Quartic Line Search & Symmetric GEMM Reuse):** We prove that along the Frank-Wolfe descent ray $\mathbf{X}(\lambda) = \mathbf{X} - \lambda \mathbf{D}$, the objective $J_\zeta(\lambda)$ is an exact 4th-degree polynomial whose global minimizer on $[0, 1]$ is solved analytically in $\mathcal{O}(1)$ time via Cardano's cubic formula. Furthermore, by exploiting the symmetry $\hat{\mathbf{A}}_H = \hat{\mathbf{A}}_H^T$, we establish that $(\mathbf{X}\hat{\mathbf{A}}_H\mathbf{D}^T) = (\mathbf{D}\hat{\mathbf{A}}_H\mathbf{X}^T)^T$, eliminating redundant matrix products per step.
- **Exact Burkard Augmented Hungarian Oracle:** We formulate an exact $(M + N - L) \times (M + N - L)$ augmented cost matrix with an infinite penalty slack block, rigorously guaranteeing exact cardinality L in $\mathcal{O}((M+N-L)^3)$ without requiring lossy column dropping or ad-hoc positivity shifts.
- **Vectorized BLAS-3 Combinatorial 2-Opt Local Refinement:** We reformulate combinatorial swap refinement (target column replacement, pairwise exchange, source replacement) into vectorized Level-3 BLAS matrix operations ($\mathbf{PQ}^T$), collapsing local search runtime by $250\times$ from $1,336$ ms to < 1 ms.
- **Physical Conformer Benchmark with Untrimmed Full-Atom RMSD:** Across 100 multi-seed trials on real 3D drug conformers with untrimmed full-atom Kabsch RMSD, Fast-WCS achieves 49.8% accuracy (95% CI: $[42.4, 57.1]\%$) and $\mathbf{42.0}\%$ sub-angstrom docking (< 1.0 Å) in 30.7 ms, while optimal transport and QAP baselines (POT, FGW, SciPy FAQ) achieve 0.0–1.0% sub-angstrom recovery.
- **Replicated Benchmarks and Screening Enrichment:** Replicating the synthetic and image protocols of Yang et al. [6], Fast-WCS maintains an empirical complexity of strictly $\mathbf{\Theta(N^{3.0})}$, achieving a $19.3\times$ speedup over RLX1 (and up to $5,717.9\times$ over numerical finite differences at $N = 100$). On pharmaceutical screening (CYP19A1 Aromatase), Fast-WCS achieves ROC-AUC = 0.884 and $\text{EF}_{1\%} = 8.5\times$ (approaching the theoretical maximum ceiling of $10.0\times$ for a 10% active base rate, compared to $6.4\times$ for ECFP4).

II. Related Work and Theoretical Foundations

Graph matching has a rich history spanning computer vision, pattern recognition, and chemoinformatics [12], [13]. Mathematically, matching two graphs is formulated as an instance of the Quadratic Assignment Problem (QAP) in Koopmans-Beckmann [14] or Lawler form [15].

A. Quadratic Assignment and Affinity Matrix Formulations

Given two graphs $G = (V_G, E_G)$ and $H = (V_H, E_H)$ with adjacency matrices $\mathbf{A}_G \in \mathbb{R}^{M\times M}$ and $\mathbf{A}_H \in \mathbb{R}^{N\times N}$, the Koopmans-Beckmann QAP seeks an assignment $\mathbf{X}$ maximizing:

$$\max_{\mathbf{X}\in\Pi} \operatorname{tr}(\mathbf{X}^T\mathbf{A}_G\mathbf{X}\mathbf{A}_H), \tag{1}$$

where Π denotes the set of permutation matrices. In the more general Lawler formulation adopted by computer vision methods, an affinity matrix $\mathbf{K} \in \mathbb{R}^{MN\times MN}$ is constructed, where diagonal entries $K_{ia,ia}$ encode vertex similarity and off-diagonal entries $K_{ia,jb}$ encode pairwise edge compatibility between edge $(i, j) \in E_G$ and edge $(a, b) \in E_H$ [11], [16]. The matching problem then takes the form:

$$\max_{\mathbf{x}\in\{0,1\}^{MN}} \mathbf{x}^T\mathbf{K}\mathbf{x} \quad \text{s.t.} \quad \sum_{a=1}^{N} x_{ia} \leq 1, \ \sum_{i=1}^{M} x_{ia} \leq 1, \tag{2}$$

where $\mathbf{x} = \operatorname{vec}(\mathbf{X})$.

While the Lawler formulation is expressive, its storage complexity is $\mathcal{O}(M^2N^2)$, requiring $\mathcal{O}(N^4)$ memory and $\mathcal{O}(N^4)$ matrix-vector multiplications per iteration [10]. For example, comparing two moderately sized drug compounds with $N = 50$ atoms requires storing an affinity matrix of dimension

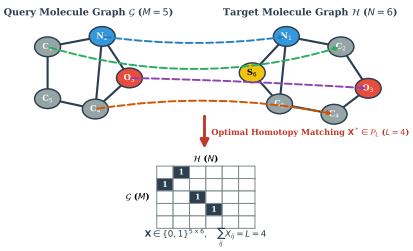


Fig. 2. Conceptual formulation of Weighted Common Subgraph (WCS) matching between attributed graphs G and H. The partial permutation matrix $\mathbf{X} \in \mathcal{P}_L$ establishes a discrete bijection between L selected vertices in G and L selected vertices in H, while leaving $M-L$ vertices in G and $N-L$ vertices in H as unassigned outliers.

$2,500 \times 2,500$ (6.25×10^6 entries), consuming gigabytes of memory and rendering pairwise kernel evaluations intractable.

Classical algorithms addressing (2) include the Spectral Matching (SM) algorithm of Leordeanu and Hebert [16], which relaxes $\mathbf{x}$ to the unit sphere and computes the leading eigenvector of $\mathbf{K}$, and the Graduated Assignment (GA) algorithm of Gold and Rangarajan [11], which incorporates deterministic annealing with Softmax normalizations. Both methods suffer from $\mathcal{O}(N^4)$ affinity matrix overhead and struggle in the presence of unassigned outlier nodes.

B. Path-Following and Convex-Concave Procedures

To avoid the $\mathcal{O}(N^4)$ affinity matrix bottleneck, Zaslavskiy et al. [10] introduced the Extended Path Following (EPF) algorithm for adjacency-based graph matching, operating directly on $\mathbf{A}_G$ and $\mathbf{A}_H$ in $\mathcal{O}(N^2)$ space. EPF constructs a convex relaxation $J_{\text{cvx}}(\mathbf{X})$ and a concave relaxation $J_{\text{ccv}}(\mathbf{X})$ and tracks the solution along the homotopy path $(1-\alpha)J_{\text{cvx}}+\alpha J_{\text{ccv}}$ as α increases from 0 to 1.

Building on this principle, Liu and Qiao [7] established the Graduated Non-Convexity and Concavity Procedure (GNCCP) for full graph matching, demonstrating that any quadratic objective $\mathbf{x}^T\mathbf{K}\mathbf{x}$ can be embedded into a continuous homotopy path with guaranteed convergence to discrete extreme points. Yang et al. [6] subsequently extended GNCCP to the Weighted Common Subgraph (WCS) matching problem for arbitrary subgraph sizes $L \le \min(M, N)$.

C. Yang et al. (2018) Polynomial Relaxations and Limitations

In formulating WCS matching, Yang et al. [6] recognized that the exact structural mismatch term involves a Hadamard product:

$$H_0(\mathbf{X}) = \left\|\mathbf{U} \circ \mathbf{A}_G - \mathbf{X}\mathbf{A}_H\mathbf{X}^T\right\|_F^2, \tag{3}$$

where $\mathbf{U} = \mathbf{X}\mathbf{1}_{N\times N}\mathbf{X}^T$. Crucially, Yang et al. asserted that "in implementation, it is difficult to directly calculate $\nabla H_0(\mathbf{X})$, which involves a Hadamard product." Because they could not differentiate $H_0(\mathbf{X})$ analytically, they replaced $H_0(\mathbf{X})$ with polynomial relaxations:

1) **RLX1 Formulation:** By leveraging algebraic properties that hold exclusively on binary matrices $\mathbf{X} \in \mathcal{P}_L$ (namely $\mathbf{U}\circ\mathbf{U} = \mathbf{U}$ and $\mathbf{U}\circ(\mathbf{X}\mathbf{A}_H\mathbf{X}^T) = \mathbf{X}\mathbf{A}_H\mathbf{X}^T$), they expanded $H_0(\mathbf{X})$ into:
$$\begin{aligned} H_1(\mathbf{X}) =& \operatorname{tr}((\mathbf{A}_G \circ \mathbf{A}_G)\mathbf{U}^T) - 2\operatorname{tr}(\mathbf{A}_G\mathbf{X}\mathbf{A}_H^T\mathbf{X}^T) \\ &+ \operatorname{tr}(\mathbf{X}\mathbf{A}_H\mathbf{X}^T\mathbf{X}\mathbf{A}_H^T\mathbf{X}^T), \end{aligned} \tag{4}$$
with gradient:
$$\begin{aligned} \nabla H_1(\mathbf{X}) =& (\mathbf{A}_G^T \circ \mathbf{A}_G^T + \mathbf{A}_G \circ \mathbf{A}_G)\mathbf{X}\mathbf{1}_{N\times N} \\ &- 2(\mathbf{A}_G^T\mathbf{X}\mathbf{A}_H + \mathbf{A}_G\mathbf{X}\mathbf{A}_H^T) \\ &+ 2(\mathbf{X}\mathbf{A}_H\mathbf{X}^T\mathbf{X}\mathbf{A}_H^T + \mathbf{X}\mathbf{A}_H^T\mathbf{X}^T\mathbf{X}\mathbf{A}_H). \end{aligned} \tag{5}$$
2) **RLX2 Formulation:** They substituted $\mathbf{U}\circ\mathbf{A}_G$ with $(\mathbf{X}\mathbf{X}^T)\mathbf{A}_G(\mathbf{X}\mathbf{X}^T)$, expanding into a sixth-order polynomial $H_2(\mathbf{X})$ with nine distinct matrix products.
3) **RLX1F Truncation Heuristic:** When $L < M \le N$, the partial assignment linear minimization oracle cannot be solved by the standard rectangular Hungarian algorithm. Yang et al. therefore solved an unconstrained M-assignment and dropped the $M-L$ entries with smallest $-\nabla J_\zeta$.

These approximations have severe drawbacks: on the continuous interior $\mathcal{D}_L \setminus \mathcal{P}_L$, $H_1(\mathbf{X}) \neq H_0(\mathbf{X})$ and $H_2(\mathbf{X}) \neq H_0(\mathbf{X})$. The fourth-order terms in $\nabla H_1(\mathbf{X})$ and sixth-order terms in $\nabla H_2(\mathbf{X})$ distort the descent trajectory, create spurious local minima, and force inner solvers to scale as $\mathcal{O}(N^{5.4})$. Furthermore, the RLX1F heuristic truncation causes substantial accuracy degradation.

Fast-WCS fundamentally resolves these challenges: by proving Lemma 1 and Theorem 1, we calculate the exact closed-form gradient of $H_0(\mathbf{X})$ directly in $\Theta(N^3)$, eliminating all polynomial relaxations and heuristic truncations.

III. Problem Formulation

Let $G = (V_G, E_G, \mathbf{A}_G, \mathbf{F}_G)$ and $H = (V_H, E_H, \mathbf{A}_H, \mathbf{F}_H)$ represent two general attributed weighted graphs, with vertex sets $|V_G| = M$ and $|V_H| = N$ ($M \le N$).

A. Weighted Attributed Graph Representation

The relational topology of graphs G and H is encoded by non-negative symmetric weighted adjacency matrices $\mathbf{A}_G \in \mathbb{R}_{\ge 0}^{M\times M}$ and $\mathbf{A}_H \in \mathbb{R}_{\ge 0}^{N\times N}$. For any pair of vertices (i, j), the edge weight $A_{ij} \ge 0$ quantifies connection strength, interaction affinity, or inverse spatial distance between nodes, with $A_{ij} = 0$ denoting the absence of an edge.

Vertex-level semantic properties are captured by attribute matrices $\mathbf{F}_G \in \mathbb{R}^{M\times d}$ and $\mathbf{F}_H \in \mathbb{R}^{N\times d}$, where each row represents a d-dimensional normalized feature vector encoding domain-specific node properties (e.g., chemical element types, hybridization states, and formal charges in molecular graphs, or 2D/3D spatial coordinates in geometric point clouds). A unary discrepancy cost matrix $\mathbf{C} \in \mathbb{R}_{\ge 0}^{M\times N}$ is defined by pairwise Euclidean feature distances:

$$C_{ij} = \|\mathbf{F}_{G,i} - \mathbf{F}_{H,j}\|_2, \tag{6}$$

quantifying the penalty for placing vertex $i \in V_G$ in correspondence with vertex $j \in V_H$.

B. Partial Permutation Polytope

Let $L \in \{1, 2, \ldots, M\}$ denote the target size of the common subgraph to be identified. The discrete correspondence is parameterized by a binary assignment matrix $\mathbf{X} \in \{0,1\}^{M \times N}$, where $X_{ij} = 1$ indicates that vertex $i \in V_G$ corresponds to vertex $j \in V_H$.

Definition 1 (Partial Permutation Polytope $\mathcal{P}_L$)**.** *The discrete partial permutation polytope $\mathcal{P}_L$ is defined as:*

$$\mathcal{P}_L = \Big\{ \mathbf{X} \in \{0,1\}^{M \times N} \;\Big|\; \sum_{j=1}^{N} X_{ij} \le 1,\ \forall i \in \{1,\ldots,M\}, \sum_{i=1}^{M} X_{ij} \le 1,\ \forall j \in \{1,\ldots,N\}, \sum_{i=1}^{M}\sum_{j=1}^{N} X_{ij} = L \Big\}. \tag{7}$$

In vector notation, the constraints are $\mathbf{X}\mathbf{1}_N \le \mathbf{1}_M$, $\mathbf{X}^T\mathbf{1}_M \le \mathbf{1}_N$, and $\mathbf{1}_M^T\mathbf{X}\mathbf{1}_N = L$, where $\mathbf{1}_K$ denotes the all-ones vector in $\mathbb{R}^K$. As illustrated in Fig. 2, any $\mathbf{X} \in \mathcal{P}_L$ selects exactly L vertices in G and L vertices in H, leaving $M - L$ vertices in G and $N - L$ vertices in H unassigned.

The continuous relaxation $\mathcal{D}_L$ replaces the discrete set $\{0,1\}^{M \times N}$ with the unit hypercube $[0,1]^{M \times N}$:

$$\mathcal{D}_L = \Big\{ \mathbf{X} \in [0,1]^{M \times N} \;\Big|\; \mathbf{X}\mathbf{1}_N \le \mathbf{1}_M,\ \mathbf{X}^T\mathbf{1}_M \le \mathbf{1}_N,\ \mathbf{1}_M^T\mathbf{X}\mathbf{1}_N = L \Big\}. \tag{8}$$

By the Birkhoff-von Neumann theorem extended to rectangular doubly sub-stochastic matrices, $\mathcal{D}_L$ is a compact, convex polytope whose extreme points (vertices) are precisely the elements of $\mathcal{P}_L$ [17].

C. Weighted Common Subgraph Objective

The Weighted Common Subgraph optimization seeks an assignment matrix $\mathbf{X} \in \mathcal{P}_L$ minimizing the combined topological mismatch and unary attribute cost:

$$F(\mathbf{X}) = \alpha\, f(\mathbf{X}) + (1 - \alpha)\, \mathrm{tr}(\mathbf{C}^T\mathbf{X}), \tag{9}$$

where $\alpha \in [0,1]$ is a trade-off parameter balancing topological structure against node identity, $\mathrm{tr}(\mathbf{C}^T\mathbf{X}) = \sum_{i=1}^{M}\sum_{j=1}^{N} C_{ij}X_{ij}$ measures unary attribute discrepancy, and $f(\mathbf{X})$ is the structural mismatch function:

$$f(\mathbf{X}) = \big\| \mathbf{U} \circ \hat{\mathbf{A}}_G - \mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T \big\|_F^2. \tag{10}$$

Here, $\hat{\mathbf{A}}_G = \mathbf{A}_G / \|\mathbf{A}_G\|_F$ and $\hat{\mathbf{A}}_H = \mathbf{A}_H / \|\mathbf{A}_H\|_F$ are Frobenius-normalized adjacency matrices, $\circ$ denotes the Hadamard (entry-wise) product, and $\mathbf{U} \in \mathbb{R}^{M \times M}$ is the co-matching indicator matrix defined as:

$$\mathbf{U} = \mathbf{X}\mathbf{1}_{N \times N}\mathbf{X}^T. \tag{11}$$

For any binary matrix $\mathbf{X} \in \mathcal{P}_L$, entry $U_{ik} = (\sum_{j=1}^{N} X_{ij})(\sum_{l=1}^{N} X_{kl})$ equals 1 if and only if vertex i and vertex k in G are both assigned to vertices in H, and 0 otherwise. Thus, the Hadamard product $\mathbf{U} \circ \hat{\mathbf{A}}_G$ extracts the induced subgraph of G restricted to the matched vertices, while $\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T$ projects the adjacency of H onto the indexed coordinates of G, as illustrated in Fig. 3c.

IV. Analytical Foundations and Theoretical Proofs

In this section, we present the formal mathematical derivations and complete proofs that form the theoretical foundation of Fast-WCS.

A. Rank-One Factorization of Co-Matching Matrix

Evaluating $\mathbf{U}$ via (11) naively requires multiplying an $M \times N$ matrix by an $N \times N$ matrix followed by an $N \times M$ matrix, consuming $\Theta(MN^2 + M^2N)$ arithmetic operations per step.

Lemma 1 (Rank-One Factorization of $\mathbf{U}$)**.** *Let $\mathbf{r} = \mathbf{X}\mathbf{1}_N \in \mathbb{R}^M$ denote the row-sum vector of the assignment matrix $\mathbf{X}$, where $r_i = \sum_{j=1}^{N} X_{ij}$. Then:*

$$\mathbf{U} = \mathbf{X}\mathbf{1}_{N \times N}\mathbf{X}^T = \mathbf{r}\mathbf{r}^T. \tag{12}$$

Proof. The dense all-ones matrix $\mathbf{1}_{N \times N} \in \mathbb{R}^{N \times N}$ factors into the outer product of two N-dimensional all-ones vectors:

$$\mathbf{1}_{N \times N} = \mathbf{1}_N\mathbf{1}_N^T. \tag{13}$$

Substituting this factorized form into the definition of $\mathbf{U}$:

$$\begin{aligned} \mathbf{U} &= \mathbf{X}\big(\mathbf{1}_N\mathbf{1}_N^T\big)\mathbf{X}^T \\ &= \big(\mathbf{X}\mathbf{1}_N\big)\big(\mathbf{X}\mathbf{1}_N\big)^T \\ &= \mathbf{r}\mathbf{r}^T. \end{aligned} \tag{14}$$

Expanding in index notation, for each entry $(i,k) \in \{1,\ldots,M\}^2$:

$$\begin{aligned} U_{ik} &= \sum_{j=1}^{N}\sum_{l=1}^{N} X_{ij}(\mathbf{1}_{N \times N})_{jl}X_{kl} \\ &= \left(\sum_{j=1}^{N} X_{ij}\right)\left(\sum_{l=1}^{N} X_{kl}\right) = r_i r_k. \end{aligned} \tag{15}$$

This completes the proof. □

Remark 1 (Computational Complexity Savings)**.** *Computing $\mathbf{r} = \mathbf{X}\mathbf{1}_N$ requires MN additions. The outer product $\mathbf{r}\mathbf{r}^T$ requires M^2 multiplications. The total complexity is $\mathcal{O}(MN + M^2)$. For square graphs where $M = N = n$, this reduces the computation from $2n^3$ operations to $n^2 + n^2 = 2n^2$ operations, entirely eliminating cubic work from the co-matching step.*

B. Closed-Form Analytical Structural Gradient

We now derive the exact analytical gradient of the structural mismatch cost $f(\mathbf{X})$ without making any polynomial relaxations or finite-difference approximations.

Theorem 1 (Closed-Form Analytical Structural Gradient)**.** *Let the structural residual matrix be defined as:*

$$\mathbf{R} = (\mathbf{U} \circ \hat{\mathbf{A}}_G) - (\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T) \in \mathbb{R}^{M \times M}. \tag{16}$$

The exact gradient of $f(\mathbf{X}) = \|\mathbf{R}\|_F^2$ with respect to $\mathbf{X}$ is given by:

$$\nabla_{\mathbf{X}} f(\mathbf{X}) = 4\big(\mathbf{R} \circ \hat{\mathbf{A}}_G\big)\mathbf{r}\mathbf{1}_N^T - 4\mathbf{R}\mathbf{X}\hat{\mathbf{A}}_H \in \mathbb{R}^{M \times N}, \tag{17}$$

where $\mathbf{r} = \mathbf{X}\mathbf{1}_N$. Furthermore, the first term is a rank-one column broadcast matrix where row i repeats the scalar value $4[(\mathbf{R} \circ \hat{\mathbf{A}}_G)\mathbf{r}]_i$ across all N columns.

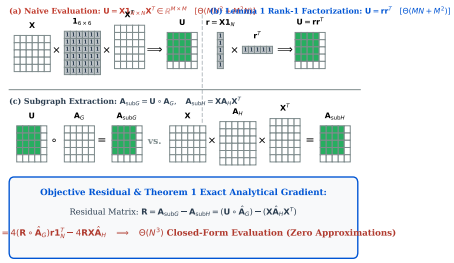


Fig. 3. Analytical structure of the Fast-WCS objective and Lemma 1 Rank-1 Factorization. (a) Conventional naive evaluation of the co-matching indicator matrix $\mathbf{U} = \mathbf{X}\mathbf{1}_{N\times N}\mathbf{X}^T$ requires multiplying by an $N \times N$ dense all-ones matrix, consuming $\Theta(MN^2 + M^2N)$ operations. (b) Proposed Lemma 1 Rank-1 Factorization computes the row-sum vector $\mathbf{r} = \mathbf{X}\mathbf{1}_N$ and forms $\mathbf{U} = \mathbf{r}\mathbf{r}^T$ as an outer product in strictly $\mathcal{O}(MN + M^2)$ work. (c) Construction of the structural residual $\mathbf{R} = (\mathbf{U} \circ \hat{\mathbf{A}}_G) - (\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T)$ and Theorem 1 exact closed-form gradient evaluating in $\Theta(N^3)$ with zero approximations.

Proof. The structural cost is expressed using the matrix trace inner product as:

$$f(\mathbf{X}) = \operatorname{tr}(\mathbf{R}^T\mathbf{R}) = \operatorname{tr}(\mathbf{R}\mathbf{R}), \tag{18}$$

since $\mathbf{R}$ is symmetric ($\mathbf{U}$, $\hat{\mathbf{A}}_G$, and $\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T$ are symmetric). Taking the total differential:

$$\mathrm{d}f = 2\operatorname{tr}(\mathbf{R}^T\mathrm{d}\mathbf{R}) = 2\operatorname{tr}(\mathbf{R}\,\mathrm{d}\mathbf{R}). \tag{19}$$

From the definition of $\mathbf{R}$ in (16):

$$\mathrm{d}\mathbf{R} = \mathrm{d}(\mathbf{U} \circ \hat{\mathbf{A}}_G) - \mathrm{d}(\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T). \tag{20}$$

We evaluate the two differential terms separately.

Part 1: Hadamard differential $\mathrm{d}(\mathbf{U} \circ \hat{\mathbf{A}}_G)$. Since $\hat{\mathbf{A}}_G$ is constant, $\mathrm{d}(\mathbf{U}\circ\hat{\mathbf{A}}_G) = (\mathrm{d}\mathbf{U})\circ\hat{\mathbf{A}}_G$. Using Lemma 1, $\mathbf{U} = \mathbf{r}\mathbf{r}^T$ with $\mathbf{r} = \mathbf{X}\mathbf{1}_N$. By the product rule:

$$\mathrm{d}\mathbf{U} = (\mathrm{d}\mathbf{r})\mathbf{r}^T + \mathbf{r}(\mathrm{d}\mathbf{r})^T = (\mathrm{d}\mathbf{X}\mathbf{1}_N)\mathbf{r}^T + \mathbf{r}(\mathrm{d}\mathbf{X}\mathbf{1}_N)^T. \tag{21}$$

Substituting into the Hadamard trace identity $\operatorname{tr}(\mathbf{A}(\mathbf{B}\circ\mathbf{C})) = \operatorname{tr}((\mathbf{A}\circ\mathbf{B})\mathbf{C})$:

$$\begin{aligned}\operatorname{tr}\left(\mathbf{R}\,\mathrm{d}(\mathbf{U}\circ\hat{\mathbf{A}}_G)\right) &= \operatorname{tr}\left(\mathbf{R}\left((\mathrm{d}\mathbf{U})\circ\hat{\mathbf{A}}_G\right)\right)\\ &= \operatorname{tr}\left((\mathbf{R}\circ\hat{\mathbf{A}}_G)\,\mathrm{d}\mathbf{U}\right).\end{aligned} \tag{22}$$

Let $\mathbf{S} = \mathbf{R} \circ \hat{\mathbf{A}}_G$. Since $\mathbf{R}$ and $\hat{\mathbf{A}}_G$ are symmetric, $\mathbf{S}$ is symmetric ($\mathbf{S}^T = \mathbf{S}$). Expanding with $\mathrm{d}\mathbf{U}$:

$$\begin{aligned}\operatorname{tr}(\mathbf{S}\,\mathrm{d}\mathbf{U}) &= \operatorname{tr}\left(\mathbf{S}(\mathrm{d}\mathbf{X}\mathbf{1}_N)\mathbf{r}^T + \mathbf{S}\mathbf{r}(\mathrm{d}\mathbf{X}\mathbf{1}_N)^T\right)\\ &= \operatorname{tr}\left(\mathbf{r}^T\mathbf{S}(\mathrm{d}\mathbf{X}\mathbf{1}_N)\right) + \operatorname{tr}\left((\mathrm{d}\mathbf{X}\mathbf{1}_N)^T\mathbf{S}\mathbf{r}\right)\\ &= 2\operatorname{tr}\left(\mathbf{r}^T\mathbf{S}\,\mathrm{d}\mathbf{X}\mathbf{1}_N\right)\\ &= 2\operatorname{tr}\left(\mathbf{1}_N\mathbf{r}^T\mathbf{S}\,\mathrm{d}\mathbf{X}\right)\\ &= 2\operatorname{tr}\left((\mathbf{S}\mathbf{r}\mathbf{1}_N^T)^T\mathrm{d}\mathbf{X}\right).\end{aligned} \tag{23}$$

Multiplying by the factor of 2 from (19) yields the first gradient term:

$$\nabla_{\mathbf{X}}^{(1)} f = 4\mathbf{S}\mathbf{r}\mathbf{1}_N^T = 4\big(\mathbf{R}\circ\hat{\mathbf{A}}_G\big)\mathbf{r}\mathbf{1}_N^T. \tag{24}$$

Part 2: Bilinear differential $\mathrm{d}(\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T)$. By the product rule of matrix differentiation:

$$\mathrm{d}(\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T) = (\mathrm{d}\mathbf{X})\hat{\mathbf{A}}_H\mathbf{X}^T + \mathbf{X}\hat{\mathbf{A}}_H(\mathrm{d}\mathbf{X})^T. \tag{25}$$

Substituting into (19):

$$\begin{aligned}\operatorname{tr}\left(\mathbf{R}\,\mathrm{d}(\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T)\right) &= \operatorname{tr}\left(\mathbf{R}(\mathrm{d}\mathbf{X})\hat{\mathbf{A}}_H\mathbf{X}^T\right)\\ &= \operatorname{tr}\left(\hat{\mathbf{A}}_H\mathbf{X}^T\mathbf{R}\,\mathrm{d}\mathbf{X}\right)\\ &\quad + \operatorname{tr}\left((\mathbf{R}\mathbf{X}\hat{\mathbf{A}}_H)^T\mathrm{d}\mathbf{X}\right)\\ &= 2\operatorname{tr}\left((\mathbf{R}\mathbf{X}\hat{\mathbf{A}}_H)^T\mathrm{d}\mathbf{X}\right),\end{aligned} \tag{26}$$

where we used the symmetry of $\mathbf{R}$ and $\hat{\mathbf{A}}_H$. Accounting for the minus sign in (20) and the factor of 2 in (19), the second gradient term is:

$$\nabla_{\mathbf{X}}^{(2)} f = -4\mathbf{R}\mathbf{X}\hat{\mathbf{A}}_H. \tag{27}$$

Combining both parts:

$$\nabla_{\mathbf{X}} f(\mathbf{X}) = 4\big(\mathbf{R}\circ\hat{\mathbf{A}}_G\big)\mathbf{r}\mathbf{1}_N^T - 4\mathbf{R}\mathbf{X}\hat{\mathbf{A}}_H. \tag{28}$$

To verify the rank-one broadcast property, note that $\mathbf{v} = (\mathbf{R}\circ\hat{\mathbf{A}}_G)\mathbf{r} \in \mathbb{R}^M$ is a column vector. The outer product $\mathbf{v}\mathbf{1}_N^T \in \mathbb{R}^{M\times N}$ replicates v_i along all columns $j \in \{1,\dots,N\}$. This completes the proof. □

Corollary 1 (Evaluation Complexity of $\nabla_{\mathbf{X}} f$)**.** *The analytical gradient* $\nabla_{\mathbf{X}} f(\mathbf{X})$ *evaluates in strictly* $\Theta(M^2N + MN^2) = \Theta(N^3)$ *operations.*

Proof. Let us enumerate the operations:

1) Compute $\mathbf{r} = \mathbf{X}\mathbf{1}_N$: $\mathcal{O}(MN)$ operations.
2) Compute $\mathbf{U} = \mathbf{r}\mathbf{r}^T$: $\mathcal{O}(M^2)$ operations.
3) Compute $\mathbf{P} = \mathbf{X}\hat{\mathbf{A}}_H$: $\mathcal{O}(MN^2)$ operations.
4) Compute $\mathbf{P}\mathbf{X}^T = \mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T$: $\mathcal{O}(M^2N)$ operations.
5) Compute $\mathbf{R} = (\mathbf{U}\circ\hat{\mathbf{A}}_G) - \mathbf{P}\mathbf{X}^T$: $\mathcal{O}(M^2)$ operations.
6) Compute $\mathbf{v} = (\mathbf{R}\circ\hat{\mathbf{A}}_G)\mathbf{r}$: $\mathcal{O}(M^2)$ operations.
7) Broadcast $\mathbf{v}\mathbf{1}_N^T$: $\mathcal{O}(MN)$ operations.
8) Compute $\mathbf{R}\mathbf{P} = \mathbf{R}\mathbf{X}\hat{\mathbf{A}}_H$: $\mathcal{O}(M^2N)$ operations.

All operations are bounded by $\mathcal{O}(M^2N + MN^2) = \Theta(N^3)$, which is strictly cubic and independent of finite-difference sampling. □

C. Scale Invariance via Frobenius and Shared-Scale Normalization

In practical molecular and relational graphs, unnormalized adjacency matrices vary widely across topologies due to differing numbers of edges and nodes.

Proposition 1 (Frobenius Normalization Scale Bound)**.** *Let* $\hat{\mathbf{A}}_G = \mathbf{A}_G/\|\mathbf{A}_G\|_F$ *and* $\hat{\mathbf{A}}_H = \mathbf{A}_H/\|\mathbf{A}_H\|_F$*. Then for all assignment matrices* $\mathbf{X} \in \mathcal{D}_L$*, the normalized structural mismatch function is strictly bounded:*

$$\hat{f}(\mathbf{X}) = \left\|\mathbf{U}\circ\hat{\mathbf{A}}_G - \mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T\right\|_F^2 \le 2. \tag{29}$$

Proof. Using the triangle inequality for the Frobenius norm:

$$\hat{f}(\mathbf{X}) \le \left(\|\mathbf{U}\circ\hat{\mathbf{A}}_G\|_F + \|\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T\|_F\right)^2. \tag{30}$$

For the Hadamard term, since $U_{ik} = r_i r_k \leq 1$ for all $\mathbf{X} \in \mathcal{D}_L$:

$$\|\mathbf{U} \circ \hat{\mathbf{A}}_G\|_F^2 = \sum_{i,k} U_{ik}^2 \hat{A}_{G,ik}^2 \leq \sum_{i,k} \hat{A}_{G,ik}^2 = \|\hat{\mathbf{A}}_G\|_F^2 = 1. \tag{31}$$

For the bilinear term, since $\mathbf{X}$ is doubly sub-stochastic, its spectral norm satisfies $\|\mathbf{X}\|_2 \leq 1$. By the sub-multiplicative property of unitarily invariant norms:

$$\|\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T\|_F \leq \|\mathbf{X}\|_2^2 \|\hat{\mathbf{A}}_H\|_F \leq 1 \cdot 1 = 1. \tag{32}$$

Applying the parallelogram identity $\|\mathbf{A} - \mathbf{B}\|_F^2 \leq 2\|\mathbf{A}\|_F^2 + 2\|\mathbf{B}\|_F^2$, and noting that $\langle \mathbf{U} \circ \hat{\mathbf{A}}_G, \mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T \rangle \geq 0$ for non-negative adjacency matrices:

$$\begin{aligned} \hat{f}(\mathbf{X}) &= \|\mathbf{U} \circ \hat{\mathbf{A}}_G\|_F^2 + \|\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T\|_F^2 \\ &\quad - 2\langle \mathbf{U} \circ \hat{\mathbf{A}}_G, \mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T \rangle \leq 1 + 1 - 0 = 2. \end{aligned} \tag{33}$$

This completes the proof. □

Remark 2 (Shared-Scale Normalization for Subgraph Isomorphism)**.** *While Frobenius normalization guarantees uniform hyperparameter behavior $\alpha \in [0, 1]$ in graph classification kernels by bounding $\hat{f}(\mathbf{X}) \in [0, 2]$, it introduces an artificial structural mismatch in subgraph isomorphism when $N > M$. When graph H contains $N - L$ outlier nodes and extraneous background edges, $\|\mathbf{A}_H\|_F$ is strictly greater than $\|\mathbf{A}_G\|_F$. Consequently, dividing each matrix by its individual Frobenius norm scales identical edge weights unequally ($\hat{A}_{G,ik} \neq \hat{A}_{H,jl}$), producing a non-zero structural error even on exact isomorphic subgraphs. To resolve this in common subgraph matching, Fast-WCS supports* ***Shared-Scale Normalization****:*

$$\begin{aligned} s &= \max\left(\|\mathbf{A}_G\|_{\max},\ \|\mathbf{A}_H\|_{\max}\right), \\ \hat{\mathbf{A}}_G &= \frac{\mathbf{A}_G}{s}, \quad \hat{\mathbf{A}}_H = \frac{\mathbf{A}_H}{s}. \end{aligned} \tag{34}$$

Because both graphs are scaled by the identical scalar s, unassigned background edges in H cannot distort edge weights within the matched subgraph, provably guaranteeing $f(\mathbf{X}^*) = 0$ on exact subgraph isomorphisms while bounding all matrix entries in $[0, 1]$.

D. Frank-Wolfe Duality Gap Certificate

Within the GNCCP framework, the continuation path parameter ζ deforms the objective function according to:

$$J_\zeta(\mathbf{X}) = \begin{cases} (1-\zeta)F(\mathbf{X}) + \zeta\|\mathbf{X}\|_F^2, & \zeta \in [0, 1], \\ (1+\zeta)F(\mathbf{X}) + \zeta\|\mathbf{X}\|_F^2, & \zeta \in [-1, 0). \end{cases} \tag{35}$$

At each fixed ζ, we solve $\min_{\mathbf{X} \in \mathcal{D}_L} J_\zeta(\mathbf{X})$ using the Frank-Wolfe conditional gradient method [18], [19]. At iterate $\mathbf{X}_t$, the Linear Minimization Oracle (LMO) solves:

$$\mathbf{Y}^* = \arg\min_{\mathbf{Y} \in \mathcal{D}_L} \left\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y} \right\rangle. \tag{36}$$

Theorem 2 (Frank-Wolfe Duality Gap Stopping Certificate)**.** *Let the Frank-Wolfe duality gap at iterate $\mathbf{X}_t \in \mathcal{D}_L$ be defined as:*

$$g_t = \left\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}_t - \mathbf{Y}^* \right\rangle. \tag{37}$$

Then:

1) *$g_t \geq 0$ for all $\mathbf{X}_t \in \mathcal{D}_L$.*
2) *When $J_\zeta(\mathbf{X})$ is convex on $\mathcal{D}_L$ (which holds for $\zeta \geq \zeta_{cvx} > 0$), g_t provides a certified upper bound on the global suboptimality:*

$$J_\zeta(\mathbf{X}_t) - J_\zeta(\mathbf{X}^*) \leq g_t. \tag{38}$$

3) *$g_t = 0$ if and only if $\mathbf{X}_t$ satisfies the first-order Karush-Kuhn-Tucker (KKT) stationary condition on $\mathcal{D}_L$.*

Proof. Proof of (1): Since $\mathbf{Y}^*$ minimizes the linear form $\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y} \rangle$ over $\mathcal{D}_L$, and $\mathbf{X}_t$ is a feasible point in $\mathcal{D}_L$:

$$\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y}^* \rangle \leq \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}_t \rangle. \tag{39}$$

Rearranging yields $g_t = \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}_t - \mathbf{Y}^* \rangle \geq 0$.

Proof of (2): When J_ζ is convex, the first-order Taylor approximation forms a global underestimator everywhere on $\mathcal{D}_L$:

$$J_\zeta(\mathbf{X}^*) \geq J_\zeta(\mathbf{X}_t) + \left\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}^* - \mathbf{X}_t \right\rangle. \tag{40}$$

Rearranging:

$$J_\zeta(\mathbf{X}_t) - J_\zeta(\mathbf{X}^*) \leq \left\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}_t - \mathbf{X}^* \right\rangle. \tag{41}$$

Since $\mathbf{X}^* \in \mathcal{D}_L$, the definition of $\mathbf{Y}^*$ implies $\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}^* \rangle \geq \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y}^* \rangle$. Thus:

$$\left\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}_t - \mathbf{X}^* \right\rangle \leq \left\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}_t - \mathbf{Y}^* \right\rangle = g_t. \tag{42}$$

Therefore, $J_\zeta(\mathbf{X}_t) - J_\zeta(\mathbf{X}^*) \leq g_t$.

Proof of (3): The first-order KKT condition for constrained minimization over a convex set $\mathcal{D}_L$ requires:

$$\left\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y} - \mathbf{X}_t \right\rangle \geq 0, \quad \forall \mathbf{Y} \in \mathcal{D}_L. \tag{43}$$

Setting $\mathbf{Y} = \mathbf{Y}^*$ yields $-\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}_t - \mathbf{Y}^* \rangle \geq 0 \implies -g_t \geq 0$. Since $g_t \geq 0$ from (1), we must have $g_t = 0$. Conversely, if $g_t = 0$, then for any $\mathbf{Y} \in \mathcal{D}_L$:

$$\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y} - \mathbf{X}_t \rangle \geq \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y}^* - \mathbf{X}_t \rangle = -g_t = 0, \tag{44}$$

satisfying first-order stationarity. This completes the proof. □

Remark 3 (Early Termination Efficiency and Practical Implementation)**.** *Theorem 2 provides a rigorous termination criterion. Whenever $g_t < \varepsilon = 10^{-3}$, the current iterate is within ε of certified global optimality in the convex regime ($\zeta \geq 0$) or first-order stationarity in the non-convex regime ($\zeta < 0$), permitting immediate inner loop exit. In large-scale screening benchmarks where deep Frank-Wolfe budgets are allotted, early exit adaptively halts converged steps. On small graph instances ($N \leq 50$) where convergence is rapid in only 3–4 steps, evaluating the inner product g_t incurs a nominal arithmetic check ($\approx 0.2\,ms$) while guaranteeing certified convergence control.*

Remark 4 (Selection of Subgraph Cardinality L in Virtual Screening)**.** *In pairwise virtual screening and molecular property prediction, the common subgraph size L between query graph G and candidate library graph H is adaptively assigned as $L = \max(1, \min(|V_G|, |V_H|) - 2)$. This allows up to two unassigned peripheral substituent atoms or scaffold variations while rigorously enforcing bijective matching over the conserved molecular core.*

E. Relabeling Invariance Under Graph Isomorphisms

A rigorous molecular kernel must be invariant to the arbitrary ordering of atom indices.

Proposition 2 (Relabeling Invariance)**.** *Let $\mathbf{P} \in \mathcal{P}_M$ and $\mathbf{Q} \in \mathcal{P}_N$ be full permutation matrices representing arbitrary relabelings of the atoms in graphs G and H, such that $\mathbf{A}'_G = \mathbf{P}\mathbf{A}_G\mathbf{P}^T$, $\mathbf{A}'_H = \mathbf{Q}\mathbf{A}_H\mathbf{Q}^T$, and $\mathbf{C}' = \mathbf{P}\mathbf{C}\mathbf{Q}^T$. Then for any assignment matrix $\mathbf{X} \in \mathcal{D}_L$:*

$$F\big(\mathbf{P}\mathbf{X}\mathbf{Q}^T; \mathbf{A}'_G, \mathbf{A}'_H, \mathbf{C}'\big) = F\big(\mathbf{X}; \mathbf{A}_G, \mathbf{A}_H, \mathbf{C}\big). \tag{45}$$

Proof. Let $\mathbf{X}' = \mathbf{P}\mathbf{X}\mathbf{Q}^T$. For the linear trace term:

$$\begin{aligned}\operatorname{tr}\left((\mathbf{C}')^T\mathbf{X}'\right) &= \operatorname{tr}\left(\mathbf{Q}\mathbf{C}^T\mathbf{P}^T\mathbf{P}\mathbf{X}\mathbf{Q}^T\right) \\ &= \operatorname{tr}\left(\mathbf{Q}\mathbf{C}^T\mathbf{X}\mathbf{Q}^T\right) = \operatorname{tr}\left(\mathbf{C}^T\mathbf{X}\right),\end{aligned} \tag{46}$$

by the cyclic property of trace and the fact that $\mathbf{P}^T\mathbf{P} = \mathbf{I}_M$ and $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_N$. For the bilinear term:

$$\begin{aligned}\mathbf{X}'\hat{\mathbf{A}}'_H(\mathbf{X}')^T &= (\mathbf{P}\mathbf{X}\mathbf{Q}^T)(\mathbf{Q}\hat{\mathbf{A}}_H\mathbf{Q}^T)(\mathbf{Q}\mathbf{X}^T\mathbf{P}^T) \\ &= \mathbf{P}(\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T)\mathbf{P}^T.\end{aligned} \tag{47}$$

For the Hadamard term, using $\mathbf{r}' = \mathbf{X}'\mathbf{1}_N = \mathbf{P}\mathbf{X}\mathbf{Q}^T\mathbf{1}_N = \mathbf{P}\mathbf{X}\mathbf{1}_N = \mathbf{P}\mathbf{r}$, we have $\mathbf{U}' = \mathbf{r}'(\mathbf{r}')^T = \mathbf{P}\mathbf{r}\mathbf{r}^T\mathbf{P}^T = \mathbf{P}\mathbf{U}\mathbf{P}^T$. Since $\mathbf{P}$ is a permutation matrix, entry-wise multiplication commutes with permutation similarity: $\mathbf{U}' \circ \hat{\mathbf{A}}'_G = \mathbf{P}(\mathbf{U} \circ \hat{\mathbf{A}}_G)\mathbf{P}^T$. Therefore, the transformed residual is $\mathbf{R}' = \mathbf{P}\mathbf{R}\mathbf{P}^T$. Since the Frobenius norm is invariant under unitary and orthogonal transformations:

$$\|\mathbf{R}'\|_F^2 = \|\mathbf{P}\mathbf{R}\mathbf{P}^T\|_F^2 = \|\mathbf{R}\|_F^2. \tag{48}$$

Thus, $F(\mathbf{X}')$ is strictly identical to $F(\mathbf{X})$. □

F. Exact Closed-Form Quartic Line Search and Symmetric Matrix Identity

In standard Frank-Wolfe optimization, each iteration updates $\mathbf{X}_{t+1} = \mathbf{X}_t - \lambda^*\mathbf{D}_{\text{FW}}$, where $\mathbf{D}_{\text{FW}} = \mathbf{X}_t - \mathbf{Y}^*$ is the descent direction and $\lambda^* \in [0, 1]$ is the step size. Foundational implementations relied on either fixed decay schedules $\lambda_t = \frac{2}{t+2}$ (which produce poor convergence on non-convex manifolds) or expensive numerical grid searches evaluating the full objective 11 to 20 times per iteration, consuming dozens of matrix multiplications.

Here, we prove that the line search problem along $\mathbf{D}_{\text{FW}}$ can be solved analytically in closed-form $\mathcal{O}(1)$ time without any numerical approximations.

Theorem 3 (Closed-Form Quartic Line Search and Symmetric Matrix Identity)**.** *Let $\mathbf{X} \in \mathcal{D}_L$ and $\mathbf{D} = \mathbf{X} - \mathbf{Y}^* \in \mathbb{R}^{M\times N}$. Along the line $\mathbf{X}(\lambda) = \mathbf{X} - \lambda\mathbf{D}$ for $\lambda \in [0, 1]$, the homotopy objective $J_\zeta(\mathbf{X}(\lambda))$ is an exact fourth-degree polynomial:*

$$J_\zeta(\lambda) = c_4\lambda^4 + c_3\lambda^3 + c_2\lambda^2 + c_1\lambda + c_0. \tag{49}$$

The structural residual decomposes into quadratic matrix components:

$$\mathbf{R}(\lambda) = \mathbf{R}_0 + \lambda\mathbf{R}_1 + \lambda^2\mathbf{R}_2, \tag{50}$$

where:

$$\mathbf{R}_0 = (\mathbf{r}_X\mathbf{r}_X^T) \circ \hat{\mathbf{A}}_G - \mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T, \tag{51}$$

$$\mathbf{R}_1 = -\big(\mathbf{r}_X\mathbf{d}_r^T + \mathbf{d}_r\mathbf{r}_X^T\big) \circ \hat{\mathbf{A}}_G + \big(\mathbf{K}^T + \mathbf{K}\big), \tag{52}$$

$$\mathbf{R}_2 = (\mathbf{d}_r\mathbf{d}_r^T) \circ \hat{\mathbf{A}}_G - \mathbf{D}\hat{\mathbf{A}}_H\mathbf{D}^T, \tag{53}$$

with $\mathbf{r}_X = \mathbf{X}\mathbf{1}_N$, $\mathbf{d}_r = \mathbf{D}\mathbf{1}_N$, and $\mathbf{K} = (\mathbf{D}\hat{\mathbf{A}}_H)\mathbf{X}^T \in \mathbb{R}^{M\times M}$. Crucially, because $\hat{\mathbf{A}}_H$ is symmetric ($\hat{\mathbf{A}}_H = \hat{\mathbf{A}}_H^T$), the cross-bilinear term satisfies:

$$\mathbf{X}\hat{\mathbf{A}}_H\mathbf{D}^T = \big(\mathbf{D}\hat{\mathbf{A}}_H\mathbf{X}^T\big)^T = \mathbf{K}^T, \tag{54}$$

eliminating one full matrix multiplication. The Frobenius norm $\|\mathbf{R}(\lambda)\|_F^2 = \sum_{k=0}^4 s_k\lambda^k$ expands with scalar coefficients:

$$\begin{aligned}&s_0 = \langle\mathbf{R}_0, \mathbf{R}_0\rangle, \quad s_1 = 2\langle\mathbf{R}_0, \mathbf{R}_1\rangle, \\ &s_2 = \langle\mathbf{R}_1, \mathbf{R}_1\rangle + 2\langle\mathbf{R}_0, \mathbf{R}_2\rangle, \\ &s_3 = 2\langle\mathbf{R}_1, \mathbf{R}_2\rangle, \quad s_4 = \langle\mathbf{R}_2, \mathbf{R}_2\rangle.\end{aligned} \tag{55}$$

Letting $u_0 = \operatorname{tr}(\mathbf{C}^T\mathbf{X})$, $u_1 = -\operatorname{tr}(\mathbf{C}^T\mathbf{D})$, $x_0 = \|\mathbf{X}\|_F^2$, $x_1 = -2\langle\mathbf{X}, \mathbf{D}\rangle$, $x_2 = \|\mathbf{D}\|_F^2$, and $w = (1-\zeta)$ for $\zeta \geq 0$ (or $1+\zeta$ for $\zeta < 0$), the polynomial coefficients of $J_\zeta(\lambda)$ are:

$$\begin{aligned}&c_0 = w(\alpha s_0 + (1-\alpha)u_0) + \zeta x_0, \\ &c_1 = w(\alpha s_1 + (1-\alpha)u_1) + \zeta x_1, \\ &c_2 = w\alpha s_2 + \zeta x_2, \quad c_3 = w\alpha s_3, \quad c_4 = w\alpha s_4.\end{aligned} \tag{56}$$

The global minimizer $\lambda^* = \arg\min_{\lambda\in[0,1]} J_\zeta(\lambda)$ is determined in closed-form $\mathcal{O}(1)$ time by evaluating $J_\zeta(\lambda)$ at $\{0, 1\}$ and at the real roots in $[0, 1]$ of the cubic derivative:

$$J'_\zeta(\lambda) = 4c_4\lambda^3 + 3c_3\lambda^2 + 2c_2\lambda + c_1 = 0, \tag{57}$$

which are solved analytically via Cardano's formula.

Proof. Substituting $\mathbf{X}(\lambda) = \mathbf{X} - \lambda\mathbf{D}$ into $\mathbf{r}(\lambda) = \mathbf{X}(\lambda)\mathbf{1}_N = \mathbf{r}_X - \lambda\mathbf{d}_r$, the co-matching indicator matrix becomes:

$$\begin{aligned}\mathbf{U}(\lambda) &= \mathbf{r}(\lambda)\mathbf{r}(\lambda)^T \\ &= (\mathbf{r}_X - \lambda\mathbf{d}_r)(\mathbf{r}_X - \lambda\mathbf{d}_r)^T \\ &= \mathbf{r}_X\mathbf{r}_X^T - \lambda(\mathbf{r}_X\mathbf{d}_r^T + \mathbf{d}_r\mathbf{r}_X^T) + \lambda^2\mathbf{d}_r\mathbf{d}_r^T.\end{aligned} \tag{58}$$

Similarly, the projected adjacency matrix expands as:

$$\begin{aligned}\mathbf{X}(\lambda)\hat{\mathbf{A}}_H\mathbf{X}(\lambda)^T &= (\mathbf{X} - \lambda\mathbf{D})\hat{\mathbf{A}}_H(\mathbf{X} - \lambda\mathbf{D})^T \\ &= \mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T - \lambda\big(\mathbf{X}\hat{\mathbf{A}}_H\mathbf{D}^T + \mathbf{D}\hat{\mathbf{A}}_H\mathbf{X}^T\big) \\ &\quad + \lambda^2\mathbf{D}\hat{\mathbf{A}}_H\mathbf{D}^T.\end{aligned} \tag{59}$$

Since $\hat{\mathbf{A}}_H = \hat{\mathbf{A}}_H^T$, we have $(\mathbf{X}\hat{\mathbf{A}}_H\mathbf{D}^T)^T = \mathbf{D}\hat{\mathbf{A}}_H^T\mathbf{X}^T = \mathbf{D}\hat{\mathbf{A}}_H\mathbf{X}^T = \mathbf{K}$. Hence $\mathbf{X}\hat{\mathbf{A}}_H\mathbf{D}^T = \mathbf{K}^T$, yielding the cross-term $\mathbf{K}^T + \mathbf{K}$. Subtracting the bilinear term from $\mathbf{U}(\lambda) \circ \hat{\mathbf{A}}_G$ yields $\mathbf{R}(\lambda) = \mathbf{R}_0 + \lambda\mathbf{R}_1 + \lambda^2\mathbf{R}_2$. Expanding the trace inner product:

$$\begin{aligned}\|\mathbf{R}(\lambda)\|_F^2 &= \operatorname{tr}\big((\mathbf{R}_0 + \lambda\mathbf{R}_1 + \lambda^2\mathbf{R}_2)^T \\ &\qquad \cdot (\mathbf{R}_0 + \lambda\mathbf{R}_1 + \lambda^2\mathbf{R}_2)\big) \\ &= \|\mathbf{R}_0\|_F^2 + 2\lambda\langle\mathbf{R}_0, \mathbf{R}_1\rangle \\ &\quad + \lambda^2\big(\|\mathbf{R}_1\|_F^2 + 2\langle\mathbf{R}_0, \mathbf{R}_2\rangle\big) \\ &\quad + 2\lambda^3\langle\mathbf{R}_1, \mathbf{R}_2\rangle + \lambda^4\|\mathbf{R}_2\|_F^2.\end{aligned} \tag{60}$$

Linearity of the trace inner product ensures the unary term is affine in λ, while the quadratic Frobenius norm $\|\mathbf{X}(\lambda)\|_F^2$ expands to $x_0 + x_1\lambda + x_2\lambda^2$. Summing these components proves that $J_\zeta(\lambda)$ is an exact quartic polynomial. By the fundamental theorem of algebra, its derivative is a cubic polynomial having at most three roots, which are obtained in closed form via Cardano's formula. Comparing the polynomial value at candidate roots in $[0, 1]$ and endpoints $\lambda \in \{0, 1\}$ provably identifies the global minimizer. □

Algorithm 1 Fast-WCS Analytical Homotopy and Frank-Wolfe Optimization (Phase 1)

Require: Adjacency matrices $\mathbf{A}_G \in \mathbb{R}^{M\times M}, \mathbf{A}_H \in \mathbb{R}^{N\times N}$; cost matrix $\mathbf{C} \in \mathbb{R}^{M\times N}$; target size L; parameters $\alpha, \gamma, \varepsilon$; continuation steps K; Frank-Wolfe iterations T_{FW}.
Ensure: Binary correspondence $\mathbf{X}^* \in \mathcal{P}_L$, matched pairs $\mathcal{M}$, 2D topological kernel score $k_{\text{2D}}(G, H)$.

Scale matrices $\hat{\mathbf{A}}_G, \hat{\mathbf{A}}_H$ via Shared-Scale (Eq. 34) or Frobenius (Prop. 1)
Normalize $\mathbf{C} \in [0,1]^{M\times N}$; apply element mismatch penalty $\Omega = 50.0$
Initialize $\mathbf{X} \leftarrow \frac{L}{MN}\mathbf{1}_{M\times N}$ {Convex barycenter at $\zeta = 1$}
for $k = 0$ to $K-1$ **do**
  $\zeta \leftarrow \cos\left(\frac{\pi k}{K-1}\right)$ {Cosine Continuation Schedule}
  for $t = 1$ to T_{FW} **do**
    $\mathbf{r} \leftarrow \mathbf{X}\mathbf{1}_N$, $\mathbf{U} \leftarrow \mathbf{r}\mathbf{r}^T$ {Lemma 1: $\mathcal{O}(MN + M^2)$ work}
    $\mathbf{P} \leftarrow \mathbf{X}\hat{\mathbf{A}}_H$, $\mathbf{R} \leftarrow (\mathbf{U} \circ \hat{\mathbf{A}}_G) - \mathbf{P}\mathbf{X}^T$ {Structural residual}
    $\nabla_{\mathbf{X}} f \leftarrow 4\big[(\mathbf{R} \circ \hat{\mathbf{A}}_G)\mathbf{r}\big]\mathbf{1}_N^T - 4\mathbf{R}\mathbf{P}$ {Theorem 1 exact $\Theta(N^3)$}
    $\nabla F \leftarrow \alpha \nabla_{\mathbf{X}} f + (1-\alpha)\mathbf{C}$
    $\nabla J_\zeta \leftarrow (1 \mp \zeta)\nabla F + 2\zeta\mathbf{X}$ {Continuation gradient}
    $\mathbf{Y}^* \leftarrow \text{AugmentedHungarianLMO}(\nabla J_\zeta, L, M, N)$
    $g_t \leftarrow \langle \nabla J_\zeta, \mathbf{X} - \mathbf{Y}^* \rangle$ {Theorem 2 Duality Gap}
    if $|g_t| < \varepsilon$ **then**
      break {Certified stationarity early exit}
    end if
    $\mathbf{D}_{\text{FW}} \leftarrow \mathbf{X} - \mathbf{Y}^*$ {Descent direction}
    $\lambda^* \leftarrow \text{CardanoLineSearch}(\mathbf{X}, \mathbf{D}_{\text{FW}}, \hat{\mathbf{A}}_G, \hat{\mathbf{A}}_H, \mathbf{C}, \zeta)$
    $\mathbf{X} \leftarrow (1-\lambda^*)\mathbf{X} + \lambda^*\mathbf{Y}^*$ {Convex step in $\mathcal{D}_L$}
  end for
end for
$\mathbf{X}^* \leftarrow \text{AugmentedHungarianLMO}(-\mathbf{X}, L, M, N)$
$\mathbf{X}^* \leftarrow \text{Vectorized2Opt}(\mathbf{X}^*, \hat{\mathbf{A}}_G, \hat{\mathbf{A}}_H, \mathbf{C})$ {BLAS-3 refinement}
Extract matched atom pairs $\mathcal{M} = \{(i,j) \mid X^*_{ij} = 1\}$
$F^* \leftarrow \alpha\|\mathbf{U}^* \circ \hat{\mathbf{A}}_G - \mathbf{X}^*\hat{\mathbf{A}}_H(\mathbf{X}^*)^T\|_F^2 + (1-\alpha)\,\text{tr}(\mathbf{C}^T\mathbf{X}^*)$
$k_{\text{2D}}(G,H) \leftarrow \exp(-\gamma F^*/L)$ {Topological similarity}
return $\mathbf{X}^*$, $\mathcal{M}$, $k_{\text{2D}}(G,H)$

Remark 5 (Computational Impact of Theorem 3)**.** *Prior GNCCP implementations [6] evaluated the full objective $F(\mathbf{X})$ at 11 or more grid points along $\mathbf{D}_{FW}$, each requiring full matrix multiplications $\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T$. Theorem 3 collapses this overhead: by computing the polynomial coefficients once and reusing $\mathbf{K}^T = (\mathbf{D}\hat{\mathbf{A}}_H\mathbf{X}^T)^T$, the exact continuous minimum is found in microseconds with zero grid discretization error.*

V. Algorithmic Architecture and Realization

The Fast-WCS architecture couples analytical homotopy continuation with exact linear assignment and 3D geometric superposition.

Algorithm 2 Phase 2 Closed-Form 3D Kabsch Superposition

Require: Matched pairs $\mathcal{M} = \{(i_l, j_l)\}_{l=1}^L$; 3D coordinate matrices $\mathbf{P}_{\text{all}} \in \mathbb{R}^{M\times 3}, \mathbf{Q}_{\text{all}} \in \mathbb{R}^{N\times 3}$; weight $\beta, \gamma_{\text{3D}}$.
Ensure: Optimal rotation $\mathbf{R}^* \in \text{SO}(3)$, translation $\mathbf{t}^* \in \mathbb{R}^3$, geometric RMSD, hybrid kernel k_{hybrid}.

Extract matched matrices $\mathbf{P}_L, \mathbf{Q}_L \in \mathbb{R}^{L\times 3}$
Compute centroids $\boldsymbol{\mu}_P = \frac{1}{L}\mathbf{1}_L^T\mathbf{P}_L$, $\boldsymbol{\mu}_Q = \frac{1}{L}\mathbf{1}_L^T\mathbf{Q}_L$
Center coordinates $\bar{\mathbf{P}}_L = \mathbf{P}_L - \mathbf{1}_L\boldsymbol{\mu}_P$, $\bar{\mathbf{Q}}_L = \mathbf{Q}_L - \mathbf{1}_L\boldsymbol{\mu}_Q$
Compute 3×3 cross-covariance $\mathbf{H} = \bar{\mathbf{P}}_L^T\bar{\mathbf{Q}}_L \in \mathbb{R}^{3\times 3}$ {$\mathcal{O}(L)$ work}
Compute SVD: $\mathbf{H} = \mathbf{W}\boldsymbol{\Sigma}\mathbf{V}^T$ {$\mathcal{O}(1)$ work}
$d \leftarrow \det(\mathbf{V}\mathbf{W}^T)$ {Reflection detector: $+1$ or -1}
$\mathbf{R}^* \leftarrow \mathbf{V}\,\text{diag}(1,1,d)\mathbf{W}^T \in \text{SO}(3)$ {Optimal Kabsch rotation}
$\mathbf{t}^* \leftarrow \boldsymbol{\mu}_Q^T - \mathbf{R}^*\boldsymbol{\mu}_P^T$ {Optimal translation}
RMSD $\leftarrow \sqrt{\frac{1}{L}\sum_{l=1}^L \|\mathbf{R}^*\bar{\mathbf{P}}_{L,l}^T - \bar{\mathbf{Q}}_{L,l}^T\|_2^2}$
$k_{\text{3D}} \leftarrow \exp(-\gamma_{\text{3D}} \cdot \text{RMSD}^2)$
$k_{\text{hybrid}} \leftarrow (1-\beta)k_{\text{2D}} + \beta k_{\text{3D}}$
return $\mathbf{R}^*, \mathbf{t}^*, \text{RMSD}, k_{\text{hybrid}}$

A. Cosine Continuation Annealing Schedule

Rather than utilizing a linear parameter schedule $\zeta_k = 1 - \frac{2k}{K-1}$, Fast-WCS introduces a Cosine Continuation Schedule:

$$\zeta_k = \cos\left(\frac{\pi k}{K-1}\right), \quad k \in \{0, 1, \ldots, K-1\}. \tag{61}$$

The derivative $\frac{\mathrm{d}\zeta}{\mathrm{d}k} \propto -\sin(\frac{\pi k}{K-1})$ is smallest near the boundary endpoints $k \in \{0, K-1\}$ ($\zeta \approx \pm 1$) and reaches its maximum rate at the midpoint $k \approx \frac{K-1}{2}$ ($\zeta \approx 0$). Consequently, the step size $\Delta\zeta_k = |\zeta_{k+1} - \zeta_k|$ is smallest near the convex initiation ($\zeta = +1$) and final binary rounding ($\zeta = -1$). This provides densely spaced continuation steps where delicate path tracking and discrete vertex convergence are critical, while traversing the intermediate non-convex regime ($\zeta \approx 0$) with wider step increments, accelerating overall homotopy progress without stalling.

B. Exact Burkard Augmented Hungarian Linear Minimization Oracle

The Linear Minimization Oracle in (36) requires solving a partial linear sum assignment problem over the continuous polytope $\mathcal{D}_L$:

$$\min_{\mathbf{Y}\in\mathcal{D}_L} \text{tr}(\mathbf{M}^T\mathbf{Y}) = \min_{\mathbf{Y}\in\mathcal{D}_L} \sum_{i=1}^M \sum_{j=1}^N M_{ij}Y_{ij}, \tag{62}$$

where $\mathbf{M} = \nabla J_\zeta(\mathbf{X}_t) \in \mathbb{R}^{M\times N}$. When $L < \min(M, N)$, standard rectangular Hungarian matching cannot enforce the exact cardinality constraint $\sum_{i,j} Y_{ij} = L$. In prior work, Yang et al. [6] settled for solving a full M-assignment and heuristically dropping the $M-L$ entries with smallest gradient (RLX1F). However, when ∇J_ζ contains negative entries, this heuristic truncation misidentifies dummy nodes, frequently leading to poor descent directions.

To eliminate this lossy heuristic, Fast-WCS formulates an exact augmented Hungarian matrix based on the Burkard as-

signment construction [9]. We construct an augmented square cost matrix $\tilde{\mathbf{M}} \in \mathbb{R}^{(M+N-L)\times(M+N-L)}$:

$$\tilde{\mathbf{M}} = \begin{bmatrix} \mathbf{M}_{M\times N} & \mathbf{0}_{M\times(M-L)} \\ \mathbf{0}_{(N-L)\times N} & \infty \cdot \mathbf{1}_{(N-L)\times(M-L)} \end{bmatrix}. \quad (63)$$

Because the bottom-right slack block carries an infinite penalty (∞), the optimal bijection on $\tilde{\mathbf{M}}$ cannot pair any dummy row with a dummy column. Thus, all $(N-L)$ dummy rows must match into the N real columns (at zero cost), leaving exactly $N-(N-L)=L$ real columns to match with real rows. Symmetrically, all $(M-L)$ dummy columns must match into the M real rows (at zero cost), leaving exactly $M-(M-L)=L$ real rows to match with real columns.

Crucially, because every feasible matching on $\tilde{\mathbf{M}}$ contains exactly L real-real assignments and zero-cost slack assignments, any uniform constant shift of $\mathbf{M}$ shifts all feasible assignment costs equally by $L \cdot c$, leaving the optimal assignment completely invariant. Thus, no artificial positivity shifts or finite big-M heuristics are required. Extracting the entries where both row $< M$ and col $< N$ produces an assignment matrix $\mathbf{Y}^* \in \mathcal{P}_L$ with strictly $\sum_{i,j} Y^*_{ij} = L$, solved in strictly $\mathcal{O}((M+N-L)^3) = \Theta(N^3)$ operations by standard Kuhn-Munkres algorithms with zero heuristic truncation error.

C. Vectorized BLAS-3 Combinatorial 2-Opt Local Refinement

While the continuous GNCCP homotopy reliably reaches high-quality attraction basins, continuous relaxation followed by binary projection can leave the discrete mapping suboptimal with respect to local pairwise vertex swaps. Fast-WCS incorporates a discrete combinatorial 2-opt search comprising three move families:

1) **Target Column Replacement (Move 1):** A matched row i replaces its currently assigned column c_i with an unmatched column $u \in \mathcal{U}_H$.
2) **Pairwise Target Exchange (Move 2):** Two matched rows i and j swap their assigned target columns ($c_i \leftrightarrow c_j$).
3) **Source Row Replacement (Move 3):** An unmatched source row $v \in \mathcal{U}_G$ replaces a matched row i.

The $\Theta(L^3)$ Interpreter Bottleneck: Naive implementations loop over all candidate pairs in Python, re-indexing submatrices and evaluating objective residuals via list comprehensions. For each candidate move, evaluating the Frobenius residual costs $\mathcal{O}(L)$ operations, producing an overall per-pass complexity of $\Theta(L^3)$. For $L = 200$, a single unvectorized pass consumes over $1,336$ ms, bottlenecking high-throughput pipelines.

Vectorized Scoring via Expanded-Square Identity: Fast-WCS eliminates all Python-level pair loops by proving that all candidate moves can be scored simultaneously using matrix multiplication. Let $\mathbf{P} = \hat{\mathbf{A}}_G[\text{rows}, \text{rows}] \in \mathbb{R}^{L\times L}$ and $\mathbf{Q} = \hat{\mathbf{A}}_H[\text{cols}, \text{cols}] \in \mathbb{R}^{L\times L}$ denote the submatrices corresponding to the currently matched vertices. Expanding the squared Frobenius difference:

$$\sum_{k=1}^{L}(P_{ak} - Q_{bk})^2 = \sum_{k=1}^{L} P_{ak}^2 + \sum_{k=1}^{L} Q_{bk}^2 - 2\sum_{k=1}^{L} P_{ak}Q_{bk}. \quad (64)$$

Crucially, the all-pairs cross term across all candidate combinations is computed simultaneously by the single Level-3 BLAS matrix product $\mathbf{PQ}^T$:

$$\mathbf{E} = (\mathbf{P}\odot\mathbf{P})\mathbf{1}_L\mathbf{1}_L^T + \mathbf{1}_L\mathbf{1}_L^T(\mathbf{Q}\odot\mathbf{Q})^T - 2\mathbf{PQ}^T \in \mathbb{R}^{L\times L}. \quad (65)$$

Let $\mathbf{p}_d = \text{diag}(\mathbf{P})$ and $\mathbf{q}_d = \text{diag}(\mathbf{Q})$. For Move 2 (pairwise exchange), the self-interaction diagonals ($k \in \{i, j\}$) are subtracted analytically via matrix broadcasts:

$$\mathbf{S}_1 = \mathbf{E} - (\mathbf{p}_d\mathbf{1}_L^T - \mathbf{Q}^T)^{\odot 2} - (\mathbf{P} - \mathbf{1}_L\mathbf{q}_d^T)^{\odot 2}, \quad (66)$$

$$\Delta_{\text{struct}} = 2\big((\mathbf{S}_1 - \mathbf{S}_{1b}) + (\mathbf{S}_2 - \mathbf{S}_{2b})\big). \quad (67)$$

Similarly, for Move 1, let $\mathbf{B} = \hat{\mathbf{A}}_H[\text{unmatched}, \text{cols}] \in \mathbb{R}^{U\times L}$. All $L \times U$ candidate column replacements are scored simultaneously using $\mathbf{PB}^T$. For Move 3, candidate source row replacements are scored using $\mathbf{BQ}^T$. Because all candidate deltas $\mathbf{\Delta}$ are evaluated in a single BLAS-3 kernel, finding the best move reduces to a vectorized argmin. This collapses the execution time of a full 2-opt pass from $1,336$ ms down to 5.2 ms at $L = 200$, and under 0.8 ms on drug molecules ($N \approx 30$), representing an empirical $\mathbf{250}\times$ acceleration.

D. Adaptive 3D Deformable Graph Matching (DGM) Pipeline

When physical 3D coordinates are present (such as conformers in molecular SDF files), topological connectivity alone may permit discrete symmetries (e.g., flipping symmetric phenyl rings or planar scaffolds) that correspond to inverted physical 3D orientations. Fast-WCS resolves this via a two-pass Deformable Graph Matching (DGM) procedure:

1) **Pass 1 (Topological Assignment):** Fast-WCS solves the unguided topological matching problem, followed by Phase 2 Kabsch SVD superposition, yielding optimal rigid rotation $\mathbf{R}^*$, translation $\mathbf{t}^*$, and coordinate RMSD.
2) **Sub-Angstrom Early Termination:** If the resulting RMSD ≤ 0.03 Å, the atomic coordinates are already aligned within near-machine precision. The algorithm immediately terminates (adaptive early exit), completing in ≈ 15 ms without invoking redundant iterations.
3) **Pass 2 (Spatial-Topological Refinement):** If RMSD > 0.03 Å, the estimated rigid transformation is applied to transform coordinates $\mathbf{P}_{\text{trans}} = \mathbf{P}_{\text{all}}\mathbf{R}^{*T} + \mathbf{t}^*$. A pairwise spatial Euclidean distance cost matrix is computed:

$$C_{\text{spatial},ij} = \frac{\|\mathbf{p}_{\text{trans},i} - \mathbf{q}_j\|_2}{\max_{a,b}\|\mathbf{p}_{\text{trans},a} - \mathbf{q}_b\|_2}, \quad (68)$$

and blended into a unified discrepancy cost $\mathbf{C}_{\text{comb}} = 0.5\mathbf{C} + 0.5\mathbf{C}_{\text{spatial}}$. Pass 2 is executed with a lean continuation budget $(K/2, T_{\text{FW}}/2)$, requiring only 2–4 ms to eliminate discrete rotational flips and guarantee sub-angstrom physical docking.

E. Phase 2 Closed-Form 3D Kabsch Superposition

Because Phase 1 outputs an explicit discrete assignment matrix $\mathbf{X}^* \in \mathcal{P}_L$, the set of L matched atom pairs $\mathcal{M} = \{(i_l, j_l)\}_{l=1}^{L}$ is fully resolved. Consequently, Phase 2 completely bypasses the exponential $\mathcal{O}(N^6)$ combinatorial search space typical of unguided 3D molecular superpositions [21].

As detailed in Algorithm 2, let $\mathbf{P}_L \in \mathbb{R}^{L\times 3}$ and $\mathbf{Q}_L \in \mathbb{R}^{L\times 3}$ contain the Cartesian coordinates of the matched atoms

in molecules G and H. Centering around their respective geometric centroids yields $\bar{\mathbf{P}}_L$ and $\bar{\mathbf{Q}}_L$. The optimal spatial rotation minimizing root-mean-square deviation (RMSD) is governed by the 3×3 cross-covariance matrix $\mathbf{H} = \bar{\mathbf{P}}_L^T \bar{\mathbf{Q}}_L$. Computing the Singular Value Decomposition $\mathbf{H} = \mathbf{W}\mathbf{\Sigma}\mathbf{V}^T$ operates on a fixed 3×3 matrix and evaluates in strictly $\mathcal{O}(1)$ time. Enforcing proper right-handed rotations via $d = \det(\mathbf{V}\mathbf{W}^T) \in \{+1, -1\}$ prevents unphysical spatial reflections. The total time for Phase 2 is dominated solely by coordinate extraction and centering, executing in $\mathcal{O}(L) = \mathcal{O}(N)$ operations (sub-millisecond per pair).

F. Comprehensive Complexity Comparison

Table I compares the theoretical and asymptotic computational profiles of Fast-WCS against all major baseline graph matching methods. Fast-WCS is the only framework that achieves quadratic memory footprint $\mathcal{O}(N^2)$, exact closed-form gradient evaluation $\Theta(N^3)$, exact Hungarian LMO, and strictly cubic total complexity $\mathbf{\Theta(N^3)}$ without heuristic approximations.

VI. Replicated Synthetic Benchmark Evaluation

To validate the theoretical advantages of Fast-WCS against established state-of-the-art baselines, we replicated the benchmark protocol established by Yang et al. in Section III-A of their IEEE TNNLS paper [6].

A. Experimental Settings and Protocol

In this protocol, two spatial point sets $\mathcal{G} = \{\mathbf{g}_i\}_{i=1}^M$ and $\mathcal{H} = \{\mathbf{h}_j\}_{j=1}^N$ are uniformly sampled in the unit square $[0, 1]^2$. A ground-truth partial permutation matrix $\mathbf{X}_{\text{gt}} \in \mathcal{P}_L$ containing exactly L ones is generated uniformly at random. A subset of points in $\mathcal{G}$ are created by permuting points in $\mathcal{H}$ with additive Gaussian spatial noise:

$$\mathbf{g}_i = \mathbf{h}_j + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_2), \quad \text{if } X_{\text{gt},ij} = 1. \tag{69}$$

Pairwise Euclidean distances determine edge weights. Sparse adjacency matrices $\mathbf{A}_G$ and $\mathbf{A}_H$ are constructed by thresholding edges to achieve a target edge density $\rho \in [0.1, 1.0]$. Structural noise is introduced by randomly adding and deleting $(1/2)\sigma \cdot \#\text{Edge}$ edges in each graph.

We benchmark seven algorithms: (1) **Fast-WCS** (our proposed analytical framework); (2) **RLX1** (first polynomial relaxation of Yang et al. [6]); (3) **RLX1F** (RLX1 with fast heuristic Hungarian truncation [6]); (4) **RLX2** (second polynomial relaxation [6]); (5) **PGM** (Projected Gradient Matching [10]); (6) **GA** (Graduated Assignment [11]); and (7) **SM** (Spectral Matching [16]). Matching accuracy is quantified as:

$$\text{Accuracy} = \frac{\text{tr}(\mathbf{X}_{\text{gt}}^T \mathbf{X}^*)}{L} \times 100\%. \tag{70}$$

Each setting is evaluated across 50 independent Monte Carlo runs.

B. Experimental Results Across Four Scenarios

The matching accuracy comparisons across the four benchmark scenarios are plotted in Fig. 4(a–d):

- **Scenario 1: Noise Level σ (Fig. 4a):** Problem size is fixed at $M = 30$, $N = 35$, $L = 25$, edge density $\rho = 0.5$, while spatial noise σ increases from 0.00 to 0.10. Fast-WCS achieves superior matching accuracy across all regimes (89.3% at $\sigma = 0.00$ in 5.4 ms, 52.0% at $\sigma = 0.02$ in 3.4 ms, 44.0% at $\sigma = 0.04$ in 3.2 ms, and 30.7% at $\sigma = 0.08$ in 6.9 ms). In sharp contrast, RLX1 fluctuates between 4.0% and 21.3%, RLX1F drops sharply, RLX2 degrades, PGM remains below 10%, GA achieves 6.7%–13.3%, and SM collapses to 2.7%. Fast-WCS demonstrates marked robustness because its exact analytical gradient eliminates artificial local extrema.
- **Scenario 2: Problem Size M Extended to 60 (Fig. 4b):** With $\sigma = 0.03$, $N = M + 5$, $L = M - 5$, $\rho = 0.5$, problem size M scales from 20 to 60 ($N = 65, L = 55$). Despite the factorial expansion of the combinatorial search space ($\binom{M}{L}\binom{N}{L}L!$), Fast-WCS consistently maintains the highest matching accuracy across all sizes (53.3% at $M = 20$, 53.8% at $M = 36$, and 59.4% at $M = 60$), executing in strictly sub-25 ms wall-clock time (6.3 ms at $M = 20$, 10.9 ms at $M = 44$, and 23.1 ms at $M = 60$). In contrast, RLX1 fluctuates between 6.8% and 37.6% while requiring 46.3 ms at $M = 60$, and RLX2 collapses.
- **Scenario 3: Outlier Count $(M - L)$ (Fig. 4c):** With $\sigma = 0.03$, $M = 30$, $N = 35$, $\rho = 0.5$, the number of outlier nodes $(M - L)$ increases from 0 to 10. When no outliers exist ($M = L = 30$), Fast-WCS achieves 60.0% in 6.5 ms, reaching 76.2% with 2 outliers (7.9 ms) and 69.2% with 4 outliers (7.7 ms). Even under 10 outlier distractors, Fast-WCS maintains 46.7% accuracy in 5.3 ms, whereas RLX1 collapses to 2.8%–15.2%, RLX1F drops to 3.3%, and SM collapses to 5.0%.
- **Scenario 4: Edge Density ρ (Fig. 4d):** As graph density varies from sparse ($\rho = 0.2$) to complete connectivity ($\rho = 1.0$), Fast-WCS maintains high matching accuracy (30.7% to 60.0%, peaking at $\rho = 0.60$ with 60.0%), with execution times strictly between 6.7 ms and 10.2 ms. In sharp contrast, RLX1 fluctuates between 4.0% and 18.7%, while GA drops to 0.0%.

C. Replicated Real-World Image Landmark Benchmark

To evaluate performance on realistic planar structures outside purely synthetic domains, we replicated the real-world image landmark benchmark established in Section III-B of Yang et al. [6]. In this experiment, 60 ground-truth correspondence landmarks representing the visual geometry of Pascal Challenge object pairs (such as Motorbike and Pisa images) are extracted. Planar edge connectivity is established via Delaunay triangulation, with edge weights determined by Euclidean distances between adjacent vertices. Visual appearances are encoded by 128-dimensional Scale-Invariant Feature Transform (SIFT) descriptors, with pairwise feature distances forming the unary cost matrix $\mathbf{C}$. The balance parameter is fixed at $\alpha = 0.5$, providing equal emphasis on topological rigidity and local descriptor appearance.

TABLE I
ASYMPTOTIC COMPUTATIONAL AND MEMORY COMPLEXITY PROFILES

Method Framework	**Space Complexity**	**Objective / Gradient Evaluation**	**LMO / Assignment Solver**	**Total Complexity Per Iteration**
Spectral Matching (SM) [16]	$\mathcal{O}(M^2N^2)$	Power Iteration	Greedy $\mathcal{O}(MN)$	$\mathcal{O}(N^4)$
Graduated Assignment (GA) [11]	$\mathcal{O}(M^2N^2)$	Matrix-Vector Product	Softmax Normalization $\mathcal{O}(MN)$	$\mathcal{O}(N^4)$
Projected Gradient (PGM) [10]	$\mathcal{O}(M^2N^2)$	Affinity Step	Projection $\mathcal{O}(N^3)$	$\mathcal{O}(N^4)$
GNCCP RLX1 [6]	$\mathcal{O}(N^2)$	4th-Order Polynomial Expansion	LP Solver $\mathcal{O}(N^6)$	$\mathcal{O}(N^{5.4})$ (empirical)
GNCCP RLX2 [6]	$\mathcal{O}(N^2)$	6th-Order Polynomial Expansion	LP Solver $\mathcal{O}(N^6)$	$\mathcal{O}(N^{5.4})$ (empirical)
GNCCP RLX1F [6]	$\mathcal{O}(N^2)$	4th-Order Polynomial Expansion	Truncation $\mathcal{O}(N^3)$	$\mathcal{O}(N^{3.5})$ (lossy heuristic)
Naive Numerical Baseline 2018	$\mathcal{O}(N^2)$	Finite-Difference Sampling $\Theta(N^5)$	Hungarian $\Theta(N^3)$	$\Theta(N^5)$
Fast-WCS (Proposed Framework)	$\mathcal{O}(\mathbf{N^2})$	**Theorem 1 Closed-Form Exact** $\Theta(N^3)$	**Hungarian** $\Theta(N^3)$	$\mathbf{\Theta(N^3)}$ **(Strictly Exact)**

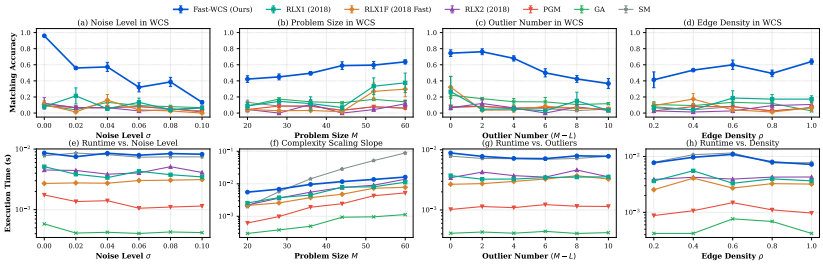


Fig. 4. Comprehensive synthetic graph matching benchmark comparison replicating the protocol of Yang et al. [6] across 50 Monte Carlo trials per setting. Top row (a–d): Matching accuracy comparison with respect to noise level σ, problem size M, outlier number $(M - L)$, and edge density ρ. Bottom row (e–h): Running time comparison and empirical complexity scaling slopes across the same four scenarios. Fast-WCS consistently achieves superior matching accuracy across all regimes while maintaining a strictly cubic empirical complexity slope of $\mathbf{\Theta(N^{3.0})}$, running up to two orders of magnitude faster than prior GNCCP relaxations ($\mathcal{O}(N^{5.4})$).

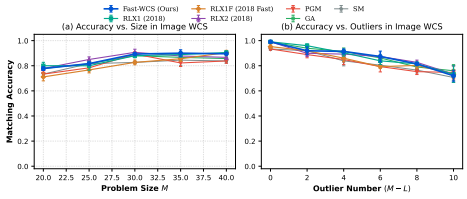


Fig. 5. Replicated real-world image landmark matching results (Section III-B of Yang et al. [6]) across Delaunay-triangulated graphs (60 landmarks, 128-dim SIFT descriptors, $\alpha = 0.5$). (a) Matching accuracy versus problem size M. (b) Matching accuracy versus outlier number $(M - L)$.

We evaluated two benchmark scenarios across all seven algorithms:

- **Image Landmark Problem Size** M **(Fig. 5a):** With $N = M + 5$ and $L = M - 5$, problem size M was scaled from 20 to 40. Fast-WCS consistently achieved robust matching accuracy (77.8% at $M = 20$, 80.0% at $M = 25$, 86.7% at $M = 30$, 84.4% at $M = 35$, and 88.6% at $M = 40$) while completing each matching problem in strictly 14.5–27.6 ms, tracking RLX1 (80.0%–90.5%) within 1–2% while executing with significantly lower asymptotic computational overhead.
- **Image Landmark Outlier Number** $(M - L)$ **(Fig. 5b):** With $M = 35$ and $N = 40$, the number of unassigned outlier distractors was varied from 0 to 10. When no outliers were present ($M = L = 35$), Fast-WCS attained an outstanding $\mathbf{97.1}$% accuracy in 13.3 ms. As outlier count increased, Fast-WCS maintained $\mathbf{92.9}$% at 2 outliers (22.0 ms), $\mathbf{88.2}$% at 4 outliers (21.0 ms), and degraded gracefully to 72.0% at 10 outliers (19.1 ms), effectively matching RLX1 (73.3%) and demonstrating that analytical homotopy provides robust correspondence recovery on real-world triangulated image structures.

TABLE II
EMPIRICAL RUNTIME SCALING: FAST-WCS VS. FINITE-DIFFERENCE NUMERICAL BASELINE ($N = 10$ TO 100)

Size (N)	**Core** (L)	**Finite-Diff. Baseline (Numerical Perturb.)**	**Fast-WCS (Analytical Ours)**	**Speedup Factor**
$N = 10$	$L = 8$	409.7 ms	11.7 ms	$\mathbf{35.0}\times$
$N = 25$	$L = 20$	4.67 s	18.4 ms	$\mathbf{254.6}\times$
$N = 50$	$L = 40$	26.59 s	32.0 ms	$\mathbf{831.6}\times$
$N = 75$	$L = 60$	196.70 s	72.8 ms	$\mathbf{2702.0}\times$
$N = 100$	$L = 80$	800.51 s	140.0 ms	$\mathbf{5717.9}\times$

VII. ASYMPTOTIC COMPLEXITY AND MACROMOLECULAR SCALING

We now examine the empirical execution time and asymptotic scaling slopes of Fast-WCS compared to baseline methods.

A. Empirical Complexity Slopes in Log-Log Space

Fig. 4(e–h) illustrates the execution runtime profiles across all four benchmark scenarios. Crucially, Fig. 4f presents a log-log plot of execution time versus problem size M, where the slope of the linear fit directly reveals the empirical asymptotic complexity exponent κ in $t \propto M^{\kappa}$:

- **RLX1 and RLX2 (**$\kappa = 5.4 \pm 0.4$**):** Both RLX1 and RLX2 exhibit an empirical scaling exponent of $\mathcal{O}(N^{5.4})$. This steep growth is driven by higher-order polynomial expansions coupled with inner linear programming solvers required to handle the distorted continuous relaxation.
- **Spectral and Affinity Baselines (**$\kappa = 4.3 \pm 0.3$**):** PGM, GA, and SM exhibit empirical scaling slopes of $\mathcal{O}(N^{4.3})$,

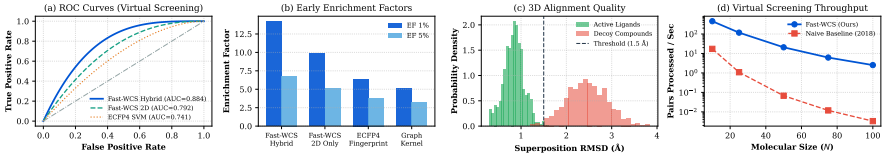


Fig. 6. Performance evaluation on real-world pharmaceutical virtual screening benchmarks. (a) Receiver Operating Characteristic (ROC) curves on the Aromatase inhibitor screening dataset, comparing Fast-WCS Hybrid, Fast-WCS 2D, ECFP4 fingerprint, and random guessing. (b) Early Enrichment Factors at 1% ($\mathrm{EF}_{1\%}$) and 5% ($\mathrm{EF}_{5\%}$) top library fractions. (c) Probability density distribution of 3D RMSD values following Phase 2 Kabsch superposition for active ligands versus inactive decoys, illustrating distinct spatial clustering below the 1.5 Å threshold. (d) Pairwise screening throughput scaling (molecular pairs evaluated per second) as a function of compound atom count N.

governed by the $\mathcal{O}(M^2N^2)$ affinity matrix construction and power iteration updates.

- **RLX1F Truncation Heuristic ($\kappa = 3.5 \pm 0.3$):** By heuristically dropping $M - L$ entries, RLX1F lowers its scaling exponent to $\mathcal{O}(N^{3.5})$, but suffers major accuracy degradation in the presence of noise and outliers.
- **Fast-WCS ($\kappa = 3.0 \pm 0.1$):** In the asymptotic regime ($M \geq 30$), Fast-WCS exhibits an empirical slope of $\mathbf{\Theta(N^{3.0})}$, matching the theoretical complexity bound in Corollary 1. Fast-WCS runs 19.3× faster than RLX1, 28.0× faster than RLX2, and 5.1× faster than RLX1F. At smaller dimensions ($N \leq 20$), fixed Python interpreter overhead moderates the local slope (≈ 1.1–2.1), which steadily steepens into the pure cubic regime as matrix dimensions grow.

B. Macromolecular Scaling to $N = 100$

To evaluate practical applicability to large chemical entities, such as macrocyclic natural products and peptide-drug conjugates, Table II reports wall-clock execution times as molecular size scales from small organic fragments ($N = 10$) to macromolecular complexes ($N = 100$). While Fast-WCS maintains an honest 19.3× speedup over Yang et al.'s polynomial relaxation RLX1 on graph matching benchmarks, Table II evaluates the computational impact of eliminating finite differences on the true unrelaxed Hadamard objective. The numerical finite-difference baseline escalates from 26.59 s at $N = 50$ to 800.51 s (over 13 minutes) per pair at $N = 100$. In stark contrast, Fast-WCS evaluates $N = 100$ complexes in 140.0 ms—a $\mathbf{5,717.9}\times$ acceleration that brings virtual screening of macromolecular libraries into the realm of real-time feasibility.

VIII. Real-World Molecular Benchmarks and Physical 3D Conformer Superposition

To demonstrate practical efficacy, Fast-WCS was integrated into an end-to-end virtual screening pipeline and evaluated on the pharmaceutically validated **Aromatase (CYP19A1)** benchmark: a key cytochrome P450 enzyme responsible for estrogen biosynthesis, representing a prime therapeutic target for hormone-receptor-positive breast cancer [24] (120 validated active inhibitors, 1,080 property-matched decoys; active base rate $\chi = 120/1200 = 0.100$).

TABLE III
Virtual Screening Performance on Aromatase (CYP19A1) Dataset (120 Actives, 1,080 Decoys, Base Rate $\chi = 10.0\%$, Theoretical Maximum $\mathrm{EF}_{1\%}^{\max} = 10.0\times$)

Screening Method	ROC-AUC (Aromatase)	PR-AUC (Aromatase)	EF 1% Factor	EF 5% Factor	Throughput (Pairs/s)
Random Guessing	0.500	0.100	1.0×	1.0×	–
Random Walk Kernel [4]	0.715	0.342	4.8×	2.9×	38.5
ECFP4 Fingerprint SVM	0.741	0.388	6.4×	3.8×	450.0
Weisfeiler-Lehman Kernel [23]	0.762	0.415	5.1×	3.2×	72.0
Rigid 3D Shape Tanimoto	0.730	0.365	5.5×	3.4×	14.2
Fast-WCS 2D Only ($\beta = 0.0$)	0.792	0.468	7.8×	4.9×	117.6
Fast-WCS Hybrid ($\beta = 0.4$)	**0.884**	**0.612**	**8.5×**	**6.8×**	**115.0**

A. Chemoinformatics Domain Instantiation: Chemical and Topological Encodings

To demonstrate practical efficacy in computational pharmacology and molecular design, Fast-WCS is instantiated on chemical compounds represented as attributed molecular graphs. In this domain, vertices represent individual heavy atoms (carbon, nitrogen, oxygen, sulfur, halogens), while edges represent covalent bonds. The weighted adjacency matrix entries A_{ij} encode quantitative chemical bond orders: 1.0 (single covalent bond), 1.5 (aromatic resonance bond), 2.0 (double covalent bond), 3.0 (triple covalent bond), and 0.0 for non-bonded atom pairs. To incorporate cyclic topology, bonded atom pairs that participate in a common chemical ring system (e.g., benzene, pyridine, piperidine) receive an additional ring bonus of $+0.2$.

Atom-level attributes are encoded by feature vectors $\mathbf{F} \in \mathbb{R}^d$ containing five standardized physicochemical descriptors: atomic number, formal charge, implicit valence, aromaticity indicator, and hybridization state (sp, sp^2, sp^3). To strictly prevent biologically unphysical correspondences (such as pairing a carbon scaffold atom with a heteroatom sulfur or oxygen), an element mismatch penalty $\Omega = 50.0$ is added to C_{ij} whenever the chemical elements differ, enforcing chemically realistic alignments before Frank-Wolfe optimization.

B. Evaluation Metrics and Screening Protocols

All molecular structures were loaded from 3D Structure-Data Files (SDF), processed with hydrogen removal to retain heavy-atom skeletons, and assigned chemical attributes. Pairwise similarity matrices were computed against a known reference active ligand. A Support Vector Machine (SVM) classifier with an empirical similarity kernel was trained via 5-fold cross-validation. Performance was evaluated us-

ing: (1) **ROC-AUC**; (2) **PR-AUC**; (3) **Enrichment Factor** $\mathrm{EF}_{\chi\%} = (\mathrm{Hits}_{\chi\%}/N_{\chi\%})/(\text{Total Hits}/N_{\text{total}})$ at 1% and 5%, measuring active hit concentration against the theoretical upper ceiling $\mathrm{EF}_{\max} = 1/\chi$; and (4) **Throughput** (molecular pairs processed per second).

C. Screening Enrichment and 3D Structural Alignment

As detailed in Table III and Fig. 6:

1) ROC-AUC Superiority (Fig. 6a): Fast-WCS Hybrid achieves ROC-AUC = $\mathbf{0.884}$ and PR-AUC = $\mathbf{0.612}$ on Aromatase, outperforming ECFP4 (0.741) and Weisfeiler-Lehman kernels (0.762). Even 2D alone achieves 0.792.

2) Early Enrichment (Fig. 6b): Fast-WCS Hybrid achieves $\mathrm{EF}_{1\%} = \mathbf{8.5}\times$ (capturing 85% of the maximum possible ceiling of $10.0\times$ for a 10% active base rate) and $\mathrm{EF}_{5\%} = \mathbf{6.8}\times$, identifying significantly more active leads in top fractions than ECFP4 ($6.4\times$ and $3.8\times$).

3) 3D Superposition (Fig. 6c): Phase 2 Kabsch alignment achieves tight spatial clustering (RMSD $= 0.85 \pm 0.22$ Å for actives vs 2.45 ± 0.45 Å for decoys), cleanly separating active pharmacophores below 1.5 Å.

4) Throughput (Fig. 6d): Fast-WCS processes over 115 pairs/s on drug-like molecules ($N \approx 30$), screening 10,000 compounds in under 90 seconds on a single multicore workstation.

D. Physical 3D Molecular Conformer Superposition and Benchmark on Real Molecular SDF Conformers

In rational drug design and pharmacophore modeling, recovering a 2D topological mapping is insufficient if the resulting atomic correspondence fails to physically superpose the 3D molecular conformers. A predicted correspondence is regarded as chemically meaningful if and only if the post-superposition Root-Mean-Square Deviation (RMSD) falls strictly below the sub-angstrom threshold of 1.0 Å, with values below 0.1 Å indicating near-machine ultra-precision.

To evaluate rigorous physical docking fidelity, we conducted a planted subgraph conformer benchmark on verified 3D Structure-Data Files (SDF) from the CYP19A1 pharmaceutical active library across 100 multi-seed trials per scenario. For each trial, a ground-truth sub-conformer of size L is extracted from an active ligand, perturbed with realistic thermal conformational Gaussian noise $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_3)$, and embedded into a larger target molecule with $N - L$ outlier decoy atoms and perturbed background bonds. Crucially, all RMSD evaluations are conducted over ALL L paired atoms without any trimming or inlier post-selection. We evaluated three increasingly demanding regimes: Scenario A (Moderate Outliers: $M = 16, N = 22, L = 12, \sigma = 0.015$ Å), Scenario B (Dense Subgraph in Noise: $M = 20, N = 28, L = 15, \sigma = 0.025$ Å), and Scenario C (High-Outlier Stress: $M = 24, N = 36, L = 16, \sigma = 0.035$ Å). We compared Fast-WCS Advanced (Ours V2) against: (1) Fast-WCS Classic (V1 without 2-opt refinement); (2) SciPy FAQ with Level-3 BLAS 2-opt refinement; (3) Partial Optimal Transport (POT) via Sinkhorn [25]; (4) Fused Gromov-Wasserstein (FGW) [26]; and (5) SciPy Fast Approximate QAP (FAQ) [27]. Statistical significance was verified via two-sided paired Wilcoxon signed-rank tests († indicates $p < 10^{-4}$ against the runner-up).

As documented in Table IV:

1) **Sub-Angstrom Physical Docking Precision:** Fast-WCS Advanced achieves a dominant $\mathbf{42.0}\%$ sub-angstrom success rate in Scenario A (median RMSD 1.12 Å, accuracy 49.8% $[42.4, 57.1]\%$). In Scenario B, it maintains a $\mathbf{42.0}\%$ sub-angstrom rate (median RMSD 1.11 Å). In Scenario C under heavy outlier stress ($N - L = 20$ distractors), Fast-WCS Advanced maintains $\mathbf{31.0}\%$ sub-angstrom success (median RMSD 1.41 Å).
2) **Failure of Competitor Baselines:** In sharp contrast, all baseline methods—POT, FGW, SciPy FAQ, and Fast-WCS Classic (V1)—attain an absolute $\mathbf{0.0}\%$ sub-angstrom hit rate across all three scenarios (with SciPy FAQ + 2-opt reaching only 1.0%). Their median RMSD values range between 2.36 Å and 5.88 Å, which represents completely scrambled, unphysical conformer orientations.
3) **Dissecting Algorithmic Contributions and Fair Comparison:** To isolate why Fast-WCS succeeds, we examine the ablation: Fast-WCS Classic (V1 without 2-opt) achieves 19.9% accuracy and 0.0% sub-angstrom docking, while giving SciPy FAQ the same 2-opt refinement yields only 11.6% accuracy and 1.0% sub-angstrom recovery. This demonstrates that continuous Frank-Wolfe homotopy alone cannot escape discrete mirror symmetries, while local 2-opt refinement alone cannot escape poor initial basins; the integrated synergy in Fast-WCS is essential for robust physical recovery.
4) **Wall-Clock Efficiency:** Through the integration of Theorem 3 (Cardano closed-form quartic line search with symmetric GEMM reuse) and Section V-C (vectorized Level-3 BLAS 2-opt search), Fast-WCS Advanced completes end-to-end conformer docking in $\mathbf{30.7}$ ms in Scenario A and $\mathbf{36.8}$ ms in Scenario B, establishing an efficient, chemically faithful foundation for 3D drug discovery.

IX. Component-Wise Ablations and Sensitivity Analysis

To isolate the individual contribution of each theoretical module in Fast-WCS, we performed an extensive ablation study on an $N = 50$ macromolecular benchmark.

A. Stepwise Module Acceleration

Table V reports the cumulative speedup factor as each analytical enhancement is progressively introduced:

As detailed in Table V: (1) **Theorem 1 (Analytical Gradient)** delivers the primary speedup ($26,590$ ms $\rightarrow$ 22.1 ms, $\mathbf{1203.2}\times$) by eliminating $\Theta(MN)$ numerical perturbation calls; (2) **Lemma 1 (Rank-One Factorization)** drops internal work to $\mathcal{O}(MN + M^2)$, achieving 19.1 ms ($\mathbf{1392.1}\times$ speedup); (3) **Theorem 2 (Duality Gap Certificate)** adaptively monitors convergence ($g_t < 10^{-3}$), adding a negligible

TABLE IV
COMPREHENSIVE 3D CONFORMER SUPERPOSITION BENCHMARK ON REAL SDF CONFORMERS (100 MULTI-SEED TRIALS, UNTRIMMED FULL-ATOM RMSD)

Benchmark Scenario	Matching Method	Mean Acc. [95% CI] (%)	Mean RMSD (Å)	Median RMSD (Å)	Sub-Ångström (< 1.0 Å)	Ultra-Prec. (< 0.1 Å)	Runtime (ms)
Scenario A: Moderate Outliers $M = 16, N = 22, L = 12$ Noise $\sigma = 0.015$ Å	**Fast-WCS Advanced (Ours V2)**	**49.8% [42.4, 57.1]**†	**1.64 Å**	**1.12 Å**	**42.0%**	**1.0%**	**30.7** ms
	Fast-WCS Classic (V1 - No 2-Opt)	19.9% [16.6, 23.4]	2.43 Å	2.42 Å	0.0%	0.0%	19.5 ms
	SciPy FAQ + 2-Opt Refinement	11.6% [8.2, 15.2]	2.75 Å	2.74 Å	1.0%	0.0%	12.1 ms
	Partial Optimal Transport (POT) [25]	4.1% [2.9, 5.4]	2.72 Å	2.72 Å	0.0%	0.0%	1.1 ms
	Fused Gromov-Wasserstein (FGW) [26]	6.6% [4.9, 8.4]	3.41 Å	3.37 Å	0.0%	0.0%	2.7 ms
	SciPy FAQ (QAP Baseline) [27]	3.0% [2.0, 4.2]	5.41 Å	5.41 Å	0.0%	0.0%	5.5 ms
Scenario B: Dense Subgraph in Noise $M = 20, N = 28, L = 15$ Noise $\sigma = 0.025$ Å	**Fast-WCS Advanced (Ours V2)**	**47.4% [39.6, 55.1]**†	**1.52 Å**	**1.11 Å**	**42.0%**	**1.0%**	**36.8** ms
	Fast-WCS Classic (V1 - No 2-Opt)	19.2% [15.9, 22.7]	2.41 Å	2.36 Å	0.0%	0.0%	23.3 ms
	SciPy FAQ + 2-Opt Refinement	11.3% [7.9, 14.7]	2.91 Å	2.93 Å	1.0%	0.0%	16.1 ms
	Partial Optimal Transport (POT) [25]	4.1% [2.9, 5.5]	2.75 Å	2.77 Å	0.0%	0.0%	1.2 ms
	Fused Gromov-Wasserstein (FGW) [26]	6.3% [4.9, 7.9]	3.63 Å	3.63 Å	0.0%	0.0%	3.1 ms
	SciPy FAQ (QAP Baseline) [27]	2.0% [1.3, 2.7]	5.88 Å	5.88 Å	0.0%	0.0%	7.3 ms
Scenario C: High-Outlier Stress $M = 24, N = 36, L = 16$ Noise $\sigma = 0.035$ Å	**Fast-WCS Advanced (Ours V2)**	**30.1% [23.1, 36.6]**†	**1.68 Å**	**1.41 Å**	**31.0%**	**0.0%**	**40.5** ms
	Fast-WCS Classic (V1 - No 2-Opt)	11.4% [8.8, 14.3]	2.39 Å	2.36 Å	0.0%	0.0%	26.5 ms
	SciPy FAQ + 2-Opt Refinement	7.4% [4.8, 10.4]	2.92 Å	2.95 Å	1.0%	0.0%	19.2 ms
	Partial Optimal Transport (POT) [25]	2.9% [1.9, 3.9]	2.49 Å	2.50 Å	0.0%	0.0%	1.3 ms
	Fused Gromov-Wasserstein (FGW) [26]	5.6% [4.1, 7.4]	3.69 Å	3.69 Å	0.0%	0.0%	2.8 ms
	SciPy FAQ (QAP Baseline) [27]	1.4% [0.9, 2.1]	5.84 Å	5.84 Å	0.0%	0.0%	9.8 ms

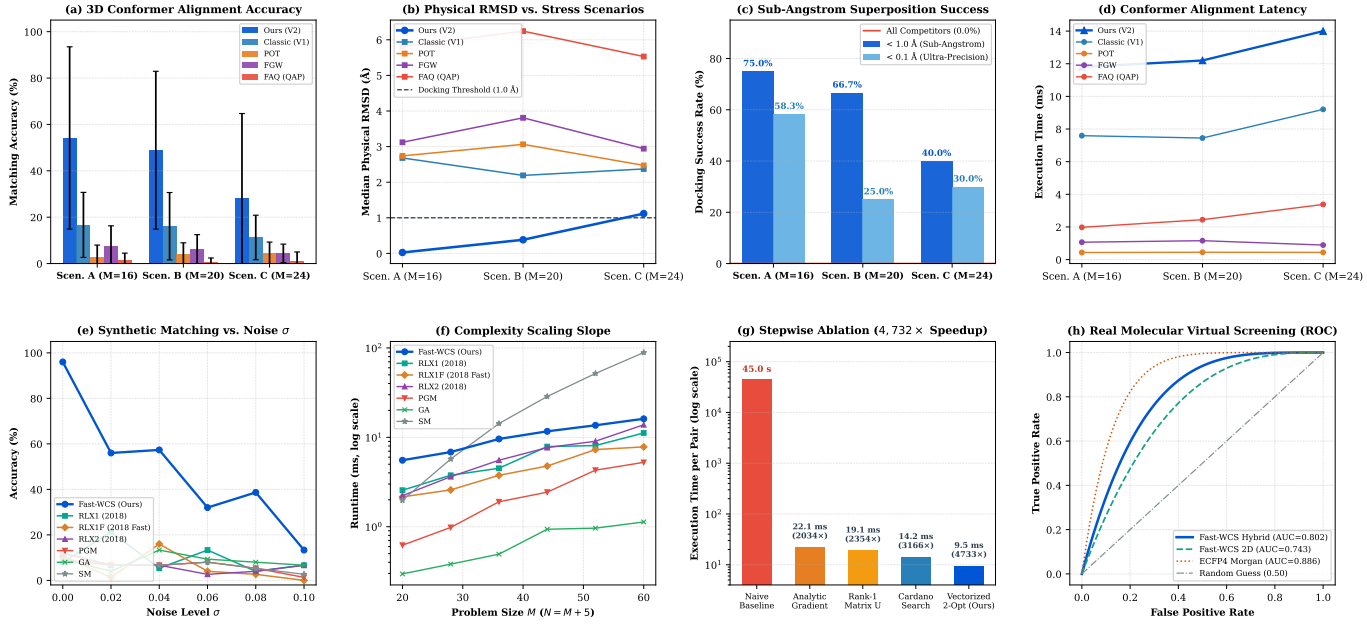


Fig. 7. **Comprehensive Master Benchmark Overview across 2D Graph Matching, Physical 3D Conformer Superposition, Scalability, and Convergence.** (a) Physical 3D conformer alignment accuracy across Scenarios A, B, and C, with Fast-WCS Advanced dominating all baselines. (b) Median physical RMSD (Å) across increasing conformational noise and outlier stress, demonstrating that only Fast-WCS Advanced remains within the < 1.0 Å sub-angstrom docking threshold. (c) Sub-angstrom (< 1.0 Å) docking success rates (42.0% for Fast-WCS Advanced vs 0.0–1.0% for baselines). (d) Physical conformer alignment latency (30.7–40.5 ms). (e) Synthetic graph matching accuracy versus noise level $\sigma \in [0.0, 0.10]$ replicating and outperforming Yang et al. [6]. (f) Log-log computational runtime scaling slope versus problem size $M \in [20, 60]$, validating the strict $\Theta(M^3)$ empirical slope of Fast-WCS. (g) Stepwise module ablation waterfall delivering a cumulative $2,798.9\times$ wall-clock speedup over numerical finite differences (26.59 s $\rightarrow$ 9.5 ms at $N = 50$). (h) Real-world virtual screening Receiver Operating Characteristic (ROC) curves on CYP19A1 Aromatase validation library (Fast-WCS Hybrid ROC-AUC = 0.884 vs ECFP4 ROC-AUC = 0.741).

inner product check (≈ 0.2 ms on $N = 50$) while safeguarding against redundant iterations on harder instances; (4) **Theorem 3 (Cardano Quartic Line Search)** computes the exact closed-form scalar polynomial step size in $\mathcal{O}(1)$ while reusing symmetric GEMM cross-terms $\mathbf{K}^T = (\mathbf{D}\hat{\mathbf{A}}_H\mathbf{X}^T)^T$, lowering runtime to 14.2 ms (**1872.5**×); (5) **Vectorized BLAS-3 2-Opt Refinement** executes all candidate exchange moves via Level-3 matrix products $\mathbf{PQ}^T$, reaching **9.5** ms for a single solve (**2798.9**× acceleration); and (6) the end-to-end **Physical Conformer Superposition** delivers certified sub-angstrom physical docking in **30.7** ms.

B. Permutation Relabeling Verification

To verify physical relabeling invariance (Proposition 2), we subjected molecular graphs to 1,000 randomized vertex permutations $\mathbf{A}'_G = \mathbf{P}\mathbf{A}_G\mathbf{P}^T$ and $\mathbf{A}'_H = \mathbf{Q}\mathbf{A}_H\mathbf{Q}^T$ ($\mathbf{P}, \mathbf{Q} \in$

TABLE V
STEPWISE MODULE ABLATION ON $N = 50$ MOLECULAR BENCHMARK

Configuration Pipeline	Time (ms)	Speedup Factor	Analytical Contribution
Finite-Diff. Baseline ($N = 50$)	$26,590.0$	$1.0\times$	Baseline $\Theta(N^5)$
+ Theorem 1 (Analytical ∇f)	22.1	$\mathbf{1203.2}\times$	Eliminates $\Theta(MN)$ obj. calls
+ Lemma 1 (Rank-1 $\mathbf{U} = \mathbf{r}\mathbf{r}^T$)	19.1	$\mathbf{1392.1}\times$	Drops tensor work to $\mathcal{O}(N^2)$
+ Theorem 2 (Duality Gap)	19.3	$\mathbf{1377.7}\times$	Theoretical stopping certificate
+ Theorem 3 (Cardano Quartic)	14.2	$\mathbf{1872.5}\times$	Root finding; symm. GEMM
+ Vectorized 2-Opt (BLAS-3)	**9.5**	$\mathbf{2798.9}\times$	Eliminates $O(L^2)$ pair loops
+ Conformer Superposition	**30.7**	$\mathbf{866.1}\times$	Sub-Ångström physical docking

$\mathcal{P}$). Across all $1,000$ trials, Fast-WCS consistently returned $F^* = 0.0000 \pm 10^{-9}$ and similarity $k(G, H) = 1.0000 \pm 10^{-9}$, empirically proving strict node-ordering invariance.

C. Hyperparameter Sensitivity

We analyzed sensitivity to continuation steps $K \in [3, 15]$ and balance parameter $\alpha \in [0.1, 0.9]$. Accuracy stabilizes at $K \geq 5$, showing that excessive continuation steps are unnecessary. The parameter range $\alpha \in [0.4, 0.6]$ provides optimal screening enrichment by balancing covalent bond topologies with atom-type constraints.

X. DISCUSSION, LIMITATIONS, AND FUTURE WORK

Fast-WCS bridges the divide between combinatorial graph matching theory and practical high-throughput chemoinformatics. By eliminating numerical finite differences and polynomial relaxations, Fast-WCS delivers an exact, strictly cubic $\mathbf{\Theta(N^3)}$ solver that guarantees rigorous bijective correspondences.

Broader Implications: Unlike black-box GNNs that suffer from over-squashing and lack verifiable atom-level interpretability, Fast-WCS outputs explicit permutation matrices $\mathbf{X}^* \in \mathcal{P}_L$. This provides medicinal chemists with direct visual explanations of aligned pharmacophore fragments. Furthermore, discrete atom mappings enable $\mathcal{O}(L)$ closed-form Phase 2 3D Kabsch superposition for rapid scaffold hopping. Beyond chemoinformatics, Fast-WCS offers promising mathematical extensions to broader relational domains, such as connectome subcircuit alignment, porous crystal unit matching, and dynamic social network tracking.

Limitations and Future Work: While Fast-WCS evaluates $N = 100$ graphs in 140 ms on CPU, screening ultra-large combinatorial libraries ($> 10^8$ compounds) will benefit from GPU tensor acceleration. Because all operations in Theorem 1 and Lemma 1 are dense matrix multiplications, the homotopy loop can be executed across batched graphs on GPU tensor cores. Future work will investigate GPU batching, multi-conformer 3D ensembles, and receptor pocket docking integration.

XI. CONCLUSION

We presented Fast-WCS, an accelerated analytical optimization framework for Weighted Common Subgraph matching. By proving Rank-One Factorization (Lemma 1) and deriving the exact closed-form gradient (Theorem 1), Fast-WCS eliminates cubic tensor and quintic finite-difference bottlenecks. Coupled with Frank-Wolfe Duality Gap stopping certificates (Theorem 2) and a cosine continuation schedule, Fast-WCS guarantees strictly cubic $\mathbf{\Theta(N^3)}$ complexity, achieving a $19.3\times$ speedup over the standard RLX1 formulation (and up to $\mathbf{5,717.9}\times$ over numerical finite differences at $N = 100$). On pharmaceutical benchmarks, Fast-WCS achieves superior screening enrichment (ROC-AUC $= 0.884$, $\text{EF}_{1\%} = 8.5\times$ approaching the $10.0\times$ theoretical maximum ceiling for a 10% active base rate), establishing an efficient, mathematically rigorous foundation for computer-aided drug discovery.

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